

Lemniscates and Newton Flow: Complete Reasoning Record

Erdős Problem #1041

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ABSTRACT

Erdős, Herzog, and Piranian ask whether every monic polynomial of degree $n \geq 2$ with zeros in the open unit disc has two zeros joined inside $\{|f| < 1\}$ by a curve of length less than 2. Every monic trinomial $f(z) = z^n + az^m + b$ with $1 \leq m < n$ and every zero in the open unit disc has the entire segment from the origin to each zero inside $\{|f| < 1\}$, so any two zeros are joined through the origin by a broken line of length below 2, in every degree and with the middle coefficient unrestricted. Separation of one critical value gives the same conclusion for an arbitrary monic polynomial: if c is a simple critical point, $v = f(c) \neq 0$, and some real centre $w_0 \in [0, 1]$ satisfies $|f(d)/v - w_0| \geq 4/3$ at every other critical point, then two zeros are joined inside $\{|f| < 1\}$ by a curve of length below 2 in every degree $n \geq 3$, provided $|v| < 1$.

The separation theorem resolves the saddle by a square-root coordinate. A disk centred at w_0 and containing the value segment $[0, 1]$ isolates the selected simple branch point. The resulting two-sheeted component is uniformised by $\zeta^2 = 1 - P$, then mapped to the unit disc so that the connector becomes a diameter. The Bergman segment inequality and the exterior-fibre capacity gap give

$$\text{length}(\Gamma)^2 \leq 2|v|^{2/n} \left(\frac{S}{n-1} \right)^{2/n} \log \frac{S^2 + S + w_0(1-w_0)}{S^2 - S + w_0(1-w_0)}.$$

The threshold $S = 4/3$ reaches every degree $n \geq 3$ and every real centre; $S = 2$ remains an immediate weaker corollary.

Write $\mu = \min_{f'(c)=0} |f(c)|$ and $\rho = \mu^{1/n}$. Two further statements hold for arbitrary monic polynomials. Every squarefree f of degree $n \geq 2$ with $\mu \leq 13/25$ has two distinct zeros joined inside $\{|f| < 1\}$ by a rectifiable curve of length strictly below 2, with no hypothesis on where the zeros lie; that proof is ordinary analysis closed by an exact rational certificate with certified quantity $X = 635762889599/1000000000000$ and closing inequality $(13/25)e^X < 1$. With the threshold removed, two zero occurrences of any monic f are joined inside $\{|f| \leq 2\mu\}$ by a path of length at most $(71/10)\rho$, and when $\mu \leq 1/2$ they are joined inside $\{|f| < 1\}$ with length at most 5.7. The same construction settles the problem outright for $\mu \leq 1/2$ whenever the first-merge component carries at least 17 zeros, and whenever its capacity defect satisfies $\text{cap}(\bar{C}) \leq (2\mu)^{1/n}/3$.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Degree three is settled in full. We also prove the sharp adjacent-zero envelope for collinear zeros, with affine Chebyshev equality configurations, and solve the primitive sparse quintic and translated cubic quotient-fibre families. An auxiliary sharp power-sum bound $\sum_{j=1}^{n-1} |f(c_j)|^{2/(n-1)} \leq (n-1)R^{2n/(n-1)}$ holds in every degree $n \geq 2$ for zeros in a closed disc of radius $R \geq 0$; the previously used exponents follow from this quadratic mean.

These results do not settle the unrestricted problem. Critical values near a tie and multiple saddles escape the separation argument, and the regime $13/25 < \mu < 1$ is untouched by the threshold argument. Exact counterexamples show that short Euclidean chords and critical spokes need not stay in the lemniscate. The remaining task is to select and control a contained connector; the critical-value budget alone does not bound its length.

Solved families and two all-degree regimes

Trinomials. Theorem 2.1 settles the Erdős–Herzog–Piranian conclusion for every monic trinomial in every degree, with a prescribed path and an unrestricted middle coefficient. The statement uses checked radial inequalities and the radius budget from the pinned Abel control-polygon source; the final conversion to one rectifiable path is ordinary mathematics (Section 2). **Separated critical values.** Theorem 6.1 bounds the length of the resolved inverse-ray connector at a simple isolated critical value, and Corollary 6.2 turns that bound into the target conclusion at separation $S = 4/3$ in every degree $n \geq 3$, for any real centre of the normalised value segment. **Regimes with no root hypothesis.** Theorem 3.1 settles every squarefree monic polynomial with least critical-value modulus at most $13/25$, in every degree. Theorem 4.1 removes the threshold and pays a constant factor, and its arity and capacity corollaries settle the problem outright on two further explicit regions. Theorem 5.1 settles degree three. **Complementary results.** Sharp solved families and the critical-value budget through degree five supply geometric and algebraic information beyond those regimes. **Open boundary.** Near-tied critical values and multiple saddles require another construction. An admissible-hub selector is one sufficient route to the target; no characterization of every possible solution is proved.

Contribution. The paper gives an all-degree path theorem for every monic trinomial, a sharp critical-value mean checked in Lean, a quantitative connector at a separated simple critical value whose threshold $S = 4/3$ reaches every degree $n \geq 3$, a short contained connector whenever the least critical-value modulus is at most $13/25$, a constant-factor connector of length at most $(71/10)\mu^{1/n}$ inside $\{|f| \leq 2\mu\}$, the full degree-three case, a sharp all-degree collinear theorem, a complete primitive sparse quintic theorem, and translated cubic quotient-fibre theorems in every degree $3q$. Lean checks exponential Newton-value decay, positive-ray collision interfaces, finite translation avoidance, and quantified root retention. Exact obstruction examples, including a Cassini counterexample to the printed global tree budget, show why those estimates cannot be promoted to a general connector theorem.

Relation to the open problem. The critical-value mean and translated cubic fibre path have complete Lean statements. The trinomial radial inequalities and radius budget are checked at their own public pin, with the final rectifiable-path interpretation ordinary. The analytic low-critical and separation arguments retain the source boundaries stated in their sections; the regime $13/25 < \mu < 1$ is untouched by the threshold argument. The checked dynamical and perturbative inputs do not construct unrestricted connectors, so the unrestricted conclusion is not established

here. A reported degree-seven counterexample announcement is a separate external claim, not a theorem of this record.

Executable review object. The selected solved-family Comparator interfaces are separate from the full proof coverage: they check the registered kernels. The complete critical-value mean and translated cubic fibre path have the source-bound Lean evidence cited in their sections. Those checks provide neither a general connector theorem nor an assessment of novelty or significance. The repository’s external-verification job compares these exact Lean propositions with separately declared challenge statements and an axiom budget, then asks Lean’s kernel to check the submitted proofs. The [formalisation manifest](#) and the commit-bound CI receipt record that check; they do not assess novelty, significance, or whether the original problem is solved.

The two selected perturbation interfaces have the following exact boundaries. For a finite type ι , an injective family of critical values $c: \iota \rightarrow \mathbb{C}$, and every $\varepsilon > 0$, the row `exists_small_translation_separating_arguments` supplies a shift of norm less than ε such that every translated value is nonzero and every two distinct translated values lie on different positive rays. Separately, for a monic split polynomial of positive degree whose roots have norm at most ρ , the row `constant_perturbation_roots_in_unitDisk` requires $\rho \geq 0, \varepsilon > 0$, the margin

$$((\text{natDegree } f + 1)\varepsilon)^{1/\text{natDegree } f} + \rho < 1,$$

and a shift of norm less than ε ; it then places every root of $f + C(\text{shift})$ in the open unit disc. These are finite perturbation and stability interfaces only: neither row supplies the missing global topology or the length-2 gluing argument.

The principal arguments and their dependencies. The trinomial proof in Section 2 uses the root equation to control an entire radial segment; it does not use Newton flow. The low-critical theorem in Section 3 follows a different chain: absence of a short connector forces separated points in a conformal disc, circle slices bound how many can occur at each radius, and the resulting root count forces too much area growth. The finite dual inequality is the summation step of that argument, not a replacement for its geometric inputs. The later solved families and obstructions can be consulted independently. Sections 9–11 distinguish the scalar Newton identity from the global topology and metric estimates that it does not supply. This record retains those distinctions when a short-paper proof refers here for detail.

1 The problem

Problem 1.1 (Erdős #1041). Let $f(z) = \prod_{i=1}^n (z - z_i)$ be monic of degree $n \geq 2$, with $z_i \in \mathbb{D}$, the open unit disc. Show that two of the roots can be joined by a curve of length less than 2 lying in the open lemniscate $E = \{z \in \mathbb{C} : |f(z)| < 1\}$.

The open-disc hypothesis is essential for this formulation. For $f(z) = z^2 - 1$, whose roots lie on the unit circle, every continuous path from -1 to 1 meets the imaginary

axis. There $|f(iy)| = 1 + y^2 \geq 1$, so no such path lies in the open unit lemniscate. Repeated roots, counted as distinct occurrences, give a constant path; the substantive case is therefore squarefree.

Numbering and current status follow Bloom’s Erdős problem catalogue [3], which records the problem as open. The original source is Problem 5 on printed p. 139 of Erdős–Herzog–Piranian [4, Problem 5, p. 139]; the preceding paragraph records the known input that one component of the lemniscate contains at least two zeros.

Recent work on polynomial lemniscates separates component counts from metric path questions. Ghosh and Ramachandran record that the open set $\{|f| < 1\}$ has $1 + \#\{j : |f(c_j)| \geq 1\}$ components, where $c_1, \dots, c_{n-1}$ are the critical points listed with multiplicity [5, Lemma 2.5]; the underlying component-wise Riemann–Hurwitz count appears in the proof of [8, Proposition 2.1]. For the binomial family $z^n - a$, the condition $|a| < 1$ therefore puts the filled unit lemniscate in the connected regime. Connectedness alone gives no path-length bound. Bishop, Eremenko and Lazebnik describe the possible shapes: every rational lemniscate is a lemniscate graph whose vertices are the critical points on the level set, each of even degree at least four [20, Definition 1.2 and Proposition 2.4], and a lemniscate graph is realised by a polynomial lemniscate up to a homeomorphism of the plane exactly when it is the boundary of its unbounded face [20, Corollary 1.6]. This topological description also gives no path-length bound. Our dynamical terminology is also standard: Sutherland calls $\dot{z} = -f(z)/f'(z)$ the Newton flow and observes that f maps its solution curves to straight lines emanating from the origin [6, p. 42]. The underlying identity $f(z(t)) = e^{-t}f(z(0))$ is a lemma of Shub, Tischler and Williams, recorded by Kozen and Stefánsson [7, Lemma 2.1], and the flow curves joining roots and critical points form the Newtonian graph, which has finitely many edges and is connected [7, §2]. The arguments below use only the value-space identity.

Two recent manuscripts are relevant. The 48-page manuscript posted by shtuka on 24 March 2026 [10, Theorem 1, p. 1] claims the unrestricted statement. Its Proposition 12 (p. 16, with proof continuing through p. 30) supplies the spanning-tree decomposition used in the final proof. The defect was located publicly in the problem’s discussion thread: on 25 March 2026 Tao observed that the invocation of Lemma 8 there is unjustified and that the flow lines need not organise into connected trees, and on 26 March 2026 the manuscript’s author agreed that the statement of Proposition 12 itself, not only its printed proof, is incorrect, and set the strategy aside. Section 11 records an independent diagnosis of the same local saddle defect, together with the Cassini obstruction to the printed global tree budget. That obstruction refutes the load-bearing metric estimate, not Erdős #1041. Pendyala’s independent June 2026 preprint [11, Thm. 1, p. 1] proves the degree-four case through a finite four-point radial lemma and a short polygonal connector. That is the degree-four result directly comparable to the root-pair problem. Together with the cubic theorem established here, it settles these two degrees; it does not supply the general-degree conclusion. The all-degree estimates below hold in every degree $n \geq 2$, under a hypothesis on the least critical-value modulus or under no root or threshold hypothesis at all, at the cost of a larger constant or a weaker containment level. The quartic theorem does not close the problem.

We study the Newton flow whose trajectories foliate the lemniscate.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no sorry, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Companion system context. The [claim and trust boundary](#), [cold-clone route to proof authority](#), and [public contribution protocol](#) are described in sibling papers. Those descriptions do not change the mathematical status of this note.

Statement	Status	Exact boundary
Erdős #1041	Open	No proof is claimed.
Every monic trinomial $z^n + az^m + b$	Checked radial bounds; ordinary path interpretation	$1 \leq m < n$ and every zero in the open unit disc; the middle coefficient is unrestricted and $ b < 1$ comes from Vieta. The pinned Abel source checks both complete spokes and the radius-sum bound. Two intermediate coefficients can break the radial mechanism.
Separated critical-value disk, $S = 4/3, n \geq 3$	Ordinary all-degree theorem; Lean-checked numerical kernel	A simple selected saddle with $ f(d)/v - w_0 \geq 4/3$ at every other critical point for some $w_0 \in [0, 1]$, and $ f(c) < 1$ for the parent conclusion. The two-sheeted uniformisation, Bergman estimate and exterior-fibre capacity gap are ordinary. No existence of such a disk is asserted.
Low critical value $\mu \leq 13/25$	Ordinary all-degree theorem with an exact rational certificate	Squarefree monic, $n \geq 2$, least critical-value modulus at most $13/25$; no hypothesis on root locations, component arity or component capacity. The analytic chain and the certificate are ordinary; nothing here is Lean-checked; the regime $13/25 < \mu < 1$ is untouched.
Unconditional constant factor $(71/10)\mu^{1/n}$	Ordinary all-degree theorem	Monic, $n \geq 2$, no further hypothesis; containment is $\{ f \leq 2\mu\}$; the target set is $\{ f < 1\}$, and the constant $71/10$ exceeds the target 2. Degree two is the separate exact case with constant 2.

Degree three	Ordinary theorem, complete for that degree	Roots in the open unit disc, listed with multiplicity; the two selected roots are distinct in the squarefree case. The spoke identity and its norm bound have a Lean companion in the research corpus, outside the pinned formal-source library.
Critical-value separation, general disk	Ordinary all-degree theorem	A simple nonzero hub, $S > \max(w_0, 1 - w_0)$ and the coefficient inequality in Theorem 6.1; the covering, Bergman and capacity arguments remain ordinary.
Critical-value budget in every degree	Lean-checked theorem	$\sum_j f(c_j) ^{1/n} \leq (n-1)R$ for every $n \geq 2$ and $R \geq 0$, with the sharper Poisson mean $\sum_j f(c_j) ^{2/(n-1)} \leq n-1$ on the closed unit disc. The complete mean is checked in <code>PaperCriticalValueMeanR10.lean</code> , with matching source-bound build evidence cited below. No inverse-ray length estimate is implied.
Coefficient energy to path length	Checked	Termwise differentiation and the length integral of the actual power series; the inverse branch and area bound must be supplied.
Uniform central free-point region	Checked	Every positive number of points; $ c_i \leq \sqrt{1 - e^{-2}}$, with no additional series hypotheses.
Newton value equation $w' = -w$	Checked local derivative	For a complex local parameter, away from critical points and along a curve tangent to $-f/f'$.
Scaled-value derivative	Checked local derivative	For the same complex local parameter, $\frac{d}{dt}(e^t f(z(t))) = 0$; constancy on a real interval is an ordinary integration step.
Ray separation of critical values	Lean-checked endpoint consumer plus ordinary finite-trajectory deduction	Lean contradicts distinct rays from an assumed exponential endpoint relation. Continuity at the endpoints and noncritical Newton evolution on the open interval yield that relation by ordinary integration; no global trajectory producer is supplied.
Ray-collision locus	Checked	$\beta = (ra - b)/(1 - r)$, $r > 0$, $r \neq 1$: one real parameter per pair.
Quartic case	Cited	Proved in [11, Thm. 1, p. 1]; does not extend to general degree.
Translated quartic quotient fibres	Ordinary theorem with a Lean-checked metric kernel	For $f(z) = P((z - h)^q)$, P monic quartic and $q \geq 2$; Pendyala supplies the quartic geometry, while Lean checks the root-lift density, exact primitive, and strict endpoint budget.

Signed-moment cyclic tetranomials	Lean-checked two-index safe-spoke theorem	For an indexed finite root family of $g(w) = w^m + aw^r + bw^s + c$, an exact signed L^2 moment budget selects two distinct indices whose complete spokes are safe; distinct root values require injectivity of the indexing map.
Concyclic zeros with $2\rho^n \leq 1$	Ordinary proof; finite exact and numerical checks	Distinct-root theorem; not Lean checked; the unrestricted concyclic case remains open.
Unrestricted proof of [10, Theorem 1, p. 1]	Printed tree budget false	Proposition 12 uses a false three-ended local saddle block, and the Cassini polynomial $z^2 - a^2$ at $a = 9/10$ refutes the printed $1/(2\pi)$ spanning-tree estimate. A different route cannot retain that coefficient. This does not refute Erdős #1041.
Constant-translation ray separation and root retention	Checked	After critical-value injectivity, arbitrary small ray avoidance and an explicit unit-disc margin.
Coefficient perturbation and slack stability	Open	Must first create injective critical values and preserve the component, collars and length budget.
Ray-separated slit-sheet decomposition	Ordinary theorem on a generic stratum	Critical points in the component are simple, critical values are nonzero, and their arguments and moduli are pairwise distinct. Simultaneous levels and the broader compact-band cut problem are not covered.
Metric fan-in	Open	The generic topology supplies a canonical inverse-ray tree but no target-scale length estimate; near-tied arguments, multiple saddles and the general compact-band gluing problem remain open.
Random search to degree 10	Numerical candidate connectors	Grid distances for sampled configurations, with no continuous sublevel certificate on any edge.

2 Trinomials

Write $E_f = \{z \in \mathbb{C} : |f(z)| < 1\}$. The first theorem gives a prescribed path between every pair of zeros of a trinomial, in every degree.

Theorem 2.1 (trinomial root connections). *Let n, m be integers with $1 \leq m < n$, and let $f(z) = z^n + az^m + b$ have every zero in $\mathbb{D}$. For every zero ζ , the segment $[0, \zeta]$ lies in E_f . Consequently any two zeros ζ_1, ζ_2 are joined in E_f by the broken line $\zeta_1 \rightarrow 0 \rightarrow \zeta_2$, of length $|\zeta_1| + |\zeta_2| < 2$.*

Proof. Vieta's formula gives $|b| < 1$. At a zero ζ the root equation $\zeta^n + a\zeta^m + b = 0$ eliminates the middle coefficient:

$$f(t\zeta) = b(1 - t^m) + \zeta^n(t^n - t^m). \quad (1)$$

For $0 \leq t < 1$ the two scalar weights $1 - t^m$ and $t^m - t^n$ are nonnegative and sum to $1 - t^n$, so

$$|f(t\zeta)| \leq |b|(1 - t^m) + |\zeta|^n(t^m - t^n) < 1 - t^n \leq 1.$$

At $t = 1$ the value is zero. Concatenating two such segments gives the length assertion. $\square$

The coefficient a carries no hypothesis; the root equation removes it before absolute values are taken. The conclusion concerns segments to zeros and makes no assertion that E_f is star-shaped.

The cancellation mechanism. For $f(z) = \sum_{k=0}^n c_k z^k$ and a zero ζ , put $S_j = \sum_{k=0}^j c_k \zeta^k$. Finite summation by parts gives

$$f(t\zeta) = \sum_{j=0}^{n-1} (t^j - t^{j+1}) S_j \quad (0 \leq t \leq 1), \quad (2)$$

since the coefficient of $c_k \zeta^k$ on the right is $t^k - t^n$ and $\sum_{k < n} c_k \zeta^k = -c_n \zeta^n$. The weights in (2) are nonnegative and sum to $1 - t^n$, so the whole radial segment is safe whenever every partial sum lies in the closed unit disc. A trinomial has only two distinct partial sums, $S_j = b$ for $j < m$ and $S_j = -\zeta^n$ for $m \leq j < n$, which is exactly the estimate above.

Two intermediate coefficients break this mechanism, and an exact example shows how.

Example 2.2 (an escaping root spoke). Let $0 < r < 1$ satisfy $r^6 > 320/327$, and set

$$f_r(z) = z^6 + \frac{1}{5}r^2z^4 - \frac{1}{5}r^4z^2 - r^6 = (z^2 - r^2)(z^4 + \frac{6}{5}r^2z^2 + r^4).$$

Every zero has modulus r : for the quartic factor put $z = rw$, so that the squared roots solve $v^2 + (6/5)v + 1 = 0$, whose two roots are complex conjugates of product one. Nevertheless

$$f_r(r/2) = -\frac{327}{320}r^6,$$

so the segment $[0, r]$ leaves E_{f_r} . This rules out a universal assertion about every origin-to-zero segment. It exhibits no counterexample to the existence of some short connection.

Evidence and exact boundary. The identity (2) is checked as [abel control polygon](#), the constant-term bound as [norm const lt one of roots lt one](#), the radial estimate as [trinomial radial norm lt one](#), and their combination with the metric budget as [trinomial erdos1041 conclusion](#). Example 2.2 is checked as [sextic guardrail](#). What the kernel proves is the pair of radial inequalities and the bound $|\zeta_1| + |\zeta_2| < 2$ on the sum of the two radii. Assembling those into a single rectifiable path object, and the passage to the one-dimensional Hausdorff measure used by the upstream statement of the problem, are ordinary steps taken here. Prior art for the trinomial conclusion was searched on 2 September 2026 and none was located; the argument is short enough that it should be assumed known until a source settles the question.

3 An unconditional regime: a small least critical value

The strongest unconditional statement in this note fixes a threshold on one scalar attached to f and asks nothing else. Throughout this section

$$\mu = \min_{f'(c)=0} |f(c)|$$

is the least critical-value modulus.

Theorem 3.1 (a small least critical value forces a short connector). *Let f be squarefree and monic of degree $n \geq 2$ with $\mu \leq 13/25$. Then two distinct roots of f are joined inside $\{|f| < 1\}$ by a rectifiable curve of length strictly below 2. No hypothesis is placed on the locations of the roots, on the number of roots in any component, or on the capacity of any component.*

Corollary 3.2 (scale-free form). *Every squarefree monic f of degree $n \geq 2$ has two distinct roots joined inside $\{|f| < (25/13)\mu\}$ by a curve of length below $2((25/13)\mu)^{1/n}$.*

Proof. Apply Theorem 3.1 to $s^{-n}f(sz)$ with $s = ((25/13)\mu)^{1/n}$, whose least critical-value modulus is $13/25$, and scale back. A repeated root occurrence gives the constant curve, so the squarefree case is the substantive one. $\square$

The failure inequalities. Assume no two distinct roots are joined inside $\{|f| < 1\}$ by a curve of length below 2. Choose a critical point at level μ and two descending inverse arcs into distinct one-root components of $\{|f| < \mu\}$; below μ each component maps conformally onto the value disc, and distinct local inverse arcs at the first critical point enter distinct components, since two inverse images of a nearby regular value in one component would contradict its degree one. Their union is a compact connected set containing two roots. Fix a regular level $t \in (\mu, 1)$, let C_t be the ancestor component of that set, let $k \geq 2$ be its root count, and put $x = \log(t/\mu)$ and $a = \text{Area}(C_t)/\pi$; Pólya's inequality [2, Theorem 1] gives $a \leq t^{2/n} < 1$. The component is a Jordan domain, and a Riemann map φ turns f/t into a finite Blaschke product of degree k , as in [8, (2.2)].

For a Riemann map the derivative has Bergman squared norm $\text{Area}(C_t)$. The Bergman kernel $K(z, w) = \pi^{-1}(1 - z\bar{w})^{-2}$ of the disc is positive on the real square, so the reproducing property, a measurable unimodular phase on $[0, r]$ and Cauchy–Schwarz give $\text{length}(\varphi([0, r]))^2 \leq \text{Area}(C_t) \int_0^r \int_0^r K(x, y) dx dy = a \log(1/(1 - r^2))$. Moving two preimages to $\pm s$ replaces the kernel integral $-\log(1 - r^2)$ by $4 \text{artanh}(s^2)$, so

$$\text{length}(\varphi([-s, s]))^2 \leq 4a \text{artanh}(s^2).$$

A pair of roots at hyperbolic distance d may be placed at $\pm \tanh(d/4)$, so a connector shorter than 2 exists as soon as $\text{artanh}(\tanh^2(d/4)) < 1/a$. Failure therefore gives

$$d(b_i, b_j) \geq D := 4 \text{artanh} \sqrt{\tanh(1/a)}, \quad \cosh(D/2) = e^{2/a}. \quad (3)$$

Failure also supplies a point $h \in C_t$ whose intrinsic distance to every root is at least 1: otherwise the relatively open sets $\{h : \text{dist}_{C_t}(h, b_j) < 1\}$ cover the connected compact set, and either two of them meet, giving a concatenated path below 2, or one of them

already contains both roots. Send h to the origin of the disc, write $d_j = d(0, b_j)$ and $\lambda(d) = -\log \tanh(d/2)$. The one-root estimate at h gives $\tanh(d_j/2) \geq \sqrt{1 - e^{-1/a}}$, and the Blaschke identity gives the level budget:

$$\lambda(d_j) \leq \frac{\delta(a)}{2}, \quad \delta(a) = -\log(1 - e^{-1/a}), \quad \sum_{j=1}^k \lambda(d_j) \geq x. \quad (4)$$

The arity floor. The consecutive-gap angular budget that these inequalities suggest is empty: placing $\lfloor k/2 \rfloor$ roots at the minimal radius and the rest at radius D beyond it, alternating, makes every consecutive law-of-cosines constraint vacuous while $\sum_j \lambda(d_j)$ grows linearly in k . A root parked radially behind another costs no angle. The constraint has to be imposed at every radius instead.

Lemma 3.3 (circle-slice packing). *Under (3), for every $r > 0$,*

$$\sum_{j=1}^k w(d_j, r) \leq \pi, \quad w(d, r) = \arccos\left(\text{clamp} \frac{\cosh d \cosh r - \cosh(D/2)}{\sinh d \sinh r}\right),$$

where clamp truncates its argument to $[-1, 1]$.

Proof. The open balls $B_j = B_{\text{hyp}}(b_j, D/2)$ are pairwise disjoint, since a common point would force $d(b_i, b_j) < D$. Fix $r > 0$. By the hyperbolic law of cosines the point of the hyperbolic circle of radius r at angle θ lies in B_j exactly when $\cosh d_j \cosh r - \sinh d_j \sinh r \cos(\theta - \theta_j) < \cosh(D/2)$, that is exactly when $|\theta - \theta_j| < w(d_j, r)$ modulo 2π . So each $B_j \cap C_r$ is an open arc of angular measure $2w(d_j, r)$, and disjoint arcs of a circle have total measure at most 2π . $\square$

Theorem 3.4 (dual arity floor). *Fix radii $r_1, \dots, r_p > 0$ and weights $\sigma_1, \dots, \sigma_p \geq 0$, put $\Sigma = \sum_i \sigma_i$ and*

$$U = \sup_{d \geq d_{\text{low}}(a)} \left[\lambda(d) - \sum_i \sigma_i w(d, r_i) \right], \quad \lambda(d_{\text{low}}(a)) = \frac{\delta(a)}{2}.$$

If $U > 0$, then failure forces $k \geq (x - \pi\Sigma)/U$.

Proof. By (4), Lemma 3.3 and the radius bound,

$$x \leq \sum_j \lambda(d_j) = \sum_j \left[\lambda(d_j) - \sum_i \sigma_i w(d_j, r_i) \right] + \sum_i \sigma_i \sum_j w(d_j, r_i) \leq kU + \pi\Sigma. \quad \square$$

Every nonnegative weight vector gives a valid floor, so the weights may be proposed by any heuristic; the certificate has to bound only Σ and U . Both $D(a)$ and $d_{\text{low}}(a)$ decrease in a , so a pair (Σ, U) certified at $a' \geq a$ remains valid at a , and k is an integer, so each real floor may be rounded up before the maximum over the available floors is taken.

From the arity floor to forced area growth. At a direction avoiding the finitely many critical-value arguments, lift the value radius from 0 to $te^{i\theta}$ from each of the k roots and split each lift at level μ . If A_0 is the total area of the one-root lobes below μ and $\varphi(z) = \sum_l b_l z^l$ maps the disc onto such a lobe, Cauchy–Schwarz in the radial variable and Parseval give

$$\text{mean}_\theta \left(\int_0^1 |\varphi'(re^{i\theta})| dr \right)^2 \leq \sum_{l \geq 1} \frac{l^2 |b_l|^2}{2l-1} \leq \sum_{l \geq 1} l |b_l|^2 = \frac{\text{Area}(\text{lobe})}{\pi},$$

so the mean total low lift length is at most $\sqrt{kA_0/\pi}$. At a regular intermediate level u , the argument principle and the coarea formula give the perimeter inequality

$$P_C(u)^2 \leq 2\pi k u A'_C(u), \quad (5)$$

because $|dz| = u d(\arg f)/|f'|$ on the level curve, its total argument variation is $2\pi k$, and Cauchy–Schwarz applies. Integrating $P_C(u)/(2\pi u)$ from μ to t bounds the mean high lift length by $\sqrt{kx(\pi a - A_0)/(2\pi)}$, so some direction has total lift length at most $M = \sqrt{ka(x+2)/2}$. Ordering the k boundary endpoints of that direction cyclically and forming k root connectors, each from two lifts and the intervening boundary arc, the whole construction lies in $\overline{C_t} \subset \{|f| < 1\}$; under failure every one of them has length at least 2, so $2k \leq 2M + P(t)$. Applying (5) at the outer level and rearranging gives the area growth inequality

$$a'(x) \geq \frac{1}{2\pi^2} \left[2\sqrt{k} - \sqrt{2a(x+2)} \right]_+^2 \quad (6)$$

on every regular interval of the ancestor, with k bounded below by Theorem 3.4. At merger levels the ancestor only gains area, which strengthens the integrated comparison. Pólya’s inequality caps a at 1 until the level reaches 1, so a trajectory forced past that cap before $t = 1$ contradicts the assumption.

The comparison is integrated cell by cell without assuming that the independently certified floors form a monotone lookup. On a cell $[x_\ell, x_r]$ with trial area m and a certified lower bound $a(x_\ell) \geq a_\ell$, a value m with $m < a_\ell + (x_r - x_\ell)g(x_\ell, x_r, m)$ is a strict lower bound for $a(x_r)$: if instead $a(x_r) \leq m$, monotonicity of the area gives $a \leq m$ throughout the cell, and integrating (6) with the nonnegative merger jumps contradicts that.

Certificate. The threshold is the level at which the forced area passes the cap. The companion program [check_erdos1041_angular_budget_closure.py](#) evaluates every accepted inequality in exact rational arithmetic, with directed rounding on every exponential, logarithm and square root, and with the rational brackets for π and $\log 2$ checked from Machin’s identity. The full replay uses 18 area-table levels, 14 radii per dual, 126 certified duals, step $1/400$, and the single initial lower area 10^{-6} . It certifies

$$X = \frac{635762889599}{1000000000000} < 0.6357629, \quad \frac{13}{25}e^X < 1,$$

which is Theorem 3.1. The quick mode certifies $X = 664373027131/1000000000000$ and the weaker threshold $51/100$. The floating optimiser that proposes the dual weights

is outside the proof; only the rational upper bounds on the proposed pairs are used. Each accepted pair is a (Σ, U) of Theorem 3.4: the quick-mode pairs at $a = 1$ are $(0.085674, 0.045865)$, $(0.126361, 0.021829)$, $(0.163157, 0.010513)$ and $(0.208928, 0.004318)$, giving $kU + \pi\Sigma$ equal to 0.4526, 0.5716, 0.6808 and 0.7945 at $k = 4, 8, 16, 32$. The supremum over d is taken by worst-first interval refinement, using that λ is decreasing and that $1 - w$ is unimodal in d with an interior minimum, so its minimum on an interval is attained at an endpoint; beyond $d = \max_i r_i + D/2$ every w vanishes and the bracket is λ alone. The closing inequality is independently checkable: with $X = 635762889599/10^{12}$ and $e^X \leq \sum_{j \leq 14} X^j/j! + (X^{15}/15!)/(1 - X/16)$, rational arithmetic gives $(13/25)e^X < 0.982000386 < 1$.

Evidence and exact boundary. The analytic chain above is ordinary mathematics. Its general inputs are the Riemann mapping theorem, the Bergman kernel, the argument principle, the coarea formula, and Pólya's area inequality $\text{Area}\{|f| \leq t\} \leq \pi t^{2/n}$ [1, printed pp. 280–282], [2, Theorem 1]. The exact rational certificate is a kernel-free replayable computation, and no part of this theorem is Lean-checked or independently reviewed. Prior art for the assembled statement is unassessed; the hyperbolic slice inequality follows from disjointness of the balls and the law of cosines and should be assumed known. The public source of record is [AngularBudgetLowCriticalClosure.md](#), which carries the full analytic chain, the certified dual table and the replay route. The exact remaining obligation is that this threshold argument does not cover $13/25 < \mu < 1$; other results in this note intersect that range, so the globally open set is not obtained by subtracting one scalar regime. Fixed-degree arguments remain stronger at $n = 4$ and $n = 5$, where the corresponding thresholds are $61/100$ and $139/250$; from $n = 6$ on the all-degree constant $13/25$ is the better statement. Erdős #1041 remains open.

4 An unconditional all-degree constant-factor path theorem

Dropping the threshold entirely costs a constant. For a monic degree- n polynomial put

$$K_t = \{z : |f(z)| \leq t\}, \quad \mu = \min_{f'(c)=0} |f(c)|, \quad \rho = \mu^{1/n}.$$

Theorem 4.1 (unconditional constant-factor path). *For every monic polynomial f of degree $n \geq 2$, two zero occurrences are joined by a possibly degenerate path of length at most*

$$\frac{71}{10} \rho$$

inside $K_{2\mu}$. If f is squarefree, their locations are distinct. If $\mu \leq 1/2$, the construction may be chosen inside $\{|f| < 1\}$ with length at most 5.7.

The constant is the rationally certified specialization of a two-parameter bound. If the selected working component contains $k \geq 2$ roots, then for every $r \in (0, 1)$ and $\lambda > 1$ the proof constructs a path in $K_{\lambda\mu}$ whose length is at most

$$\sqrt{\frac{2}{k}} \left(\frac{\sqrt{2}r}{(1-r)^2} + \lambda^{1/n} \left(\sqrt{\log(\lambda/r)} + \frac{\pi}{\sqrt{\log \lambda}} \right) \right) \rho. \quad (\text{CF})$$

For $n \geq 3$, the choice $\lambda = 2, r = 3/20$ makes the bracket less than $71/10$: the largest case is $n = 3$, where exact rational bounds give $66517563/9392500 < 71/10$. Degree two has the exact root-segment bound 2ρ .

Proof architecture. If f has a repeated zero, the constant path between two occurrences proves the theorem. Otherwise f is squarefree and $\mu > 0$. Choose a critical point at level μ and a component at a regular level $t \in [\mu, \lambda\mu]$ that contains the corresponding first merge. If a regular component C' at level σ contains k' roots, winding and Cauchy–Schwarz give the local perimeter estimate

$$\mathcal{H}^1(\partial C')^2 \leq 2\pi k' \sigma \frac{d}{d\sigma} \text{Area}(K_\sigma \cap C'). \quad (\text{CF1})$$

Pólya’s area inequality $\text{Area}(K_T) \leq \pi T^{2/n}$ [2, Theorem 1] then selects t with a degree-free boundary-length bound.

For an argument avoiding the finitely many critical-value rays, lift the radial value segment from each root to ∂C_t . Split each lift at level $r\mu$. Koebe distortion controls the low pieces. The exact precritical conformal-radius energy inequality

$$\sum_i \frac{\mu^2}{|f'(z_i)|^2} \leq \rho^2 \quad (\text{CF2})$$

aggregates those pieces without paying once per root. Coarea and (CF1) control the high pieces on average over the argument. Choose one argument realising that average, order the lift endpoints cyclically on ∂C_t , and use the shortest adjacent boundary arc. Averaging the two lifts and the boundary arcs over the k adjacent pairs gives (CF). The three contributions are, explicitly,

$$\begin{aligned} \frac{2}{k} \sum_i \ell_i^{\text{low}} &\leq \frac{2r\rho}{\sqrt{k}(1-r)^2}, \\ \frac{2}{k} \sum_i \ell_i^{\text{high}} &\leq \sqrt{\frac{2}{k}} \sqrt{\log(\lambda/r)} T^{1/n}, \\ \frac{\mathcal{H}^1(\partial C_t)}{k} &\leq \sqrt{\frac{2}{k}} \frac{\pi T^{1/n}}{\sqrt{\log \lambda}}. \end{aligned}$$

The low bound holds on every ray, the high bound on the ray chosen from its mean, and the boundary bound at the level selected before either choice. They therefore hold together. Adding and substituting $T^{1/n} = \lambda^{1/n} \rho$ recovers (CF). For the final assertion, the repeated-zero case is already a constant path, and in degree two the root segment has length $2\rho = 2\sqrt{\mu} \leq \sqrt{2}$. For $n \geq 3$, take $\lambda = 2$ and $r = 3/20$. Since $k \geq 2$ and $\mu \leq 1/2$, (CF) gives

$$\begin{aligned} \text{length} &\leq \frac{60\sqrt{2}}{289} \mu^{1/n} + (2\mu)^{1/n} \left(\sqrt{\log(40/3)} + \frac{\pi}{\sqrt{\log 2}} \right) \\ &\leq \frac{60\sqrt{2}}{289} + \sqrt{\log(40/3)} + \frac{\pi}{\sqrt{\log 2}} \\ &= 5.676476 \dots < 5.7. \end{aligned}$$

Here $2\mu \leq 1$, and the mean-value choice may be taken at a regular level strictly below the top of its positive-measure window. Thus the entire path lies in $\{|f| < 1\}$. $\square$

The bracket (CF) retains two quantities that the constant 71/10 discards, namely the root count k of the working component through the factor $\sqrt{2/k}$, and the capacity of that component through the area input. Keeping either one turns the constant-factor theorem into the target conclusion on an explicit region.

Corollary 4.2 (high-arity closure). *Let f be monic with every root in the open unit disc, let c_* be a critical point with $|f(c_*)| = \mu$, and let k_0 be the number of roots, counted with multiplicity, in the component of K_μ containing c_* . Then Erdős #1041 holds for f in each of the three cases*

$$\mu \leq \frac{1}{2} \text{ and } k_0 \geq 17, \quad \mu \leq \frac{1}{4} \text{ and } k_0 \geq 12, \quad \mu \leq \frac{1}{8} \text{ and } k_0 \geq 10.$$

Proof. Every working component C_t in the proof of Theorem 4.1 contains the first-merge component, so $k \geq k_0$. For the first case take $\lambda = 2$, $r = 13/100$; then $(2\mu)^{1/n} \leq 1$ and $\rho \leq 1$, and the exact bounds $\sqrt{2} < 283/200$, $\sqrt{\log(200/13)} < 5/3$ and $\pi/\sqrt{\log 2} < (22/7)/(104/125) = 1375/364$ make the bracket in (CF) at most $15668813/2755116$, whose square is $34 - 12570881068535/7590664173456 < 34$. Hence $k_0 \geq 17$ gives squared length below $(2/17) \cdot 34 = 4$. For the second case take $\lambda = 4$, $r = 3/25$: the bracket is below $6075221/1273888$, whose square is $24 - 2038665078215/1622790636544 < 24$, and $2/k_0 \leq 1/6$ gives squared length below 4. For the third take $\lambda = 8$, $r = 11/100$: the bracket is below $55629121/12475575$, whose square is $20 - 18200328379859/155639971580625 < 20$, and $2/k_0 \leq 1/5$ again gives squared length below 4. In each case $\lambda\mu \leq 1$, so Lemma-level freedom in the choice of the regular level t keeps $t < 1$ and the containment strict. $\square$

Corollary 4.3 (capacity-defect closure). *Keep the hypotheses of Corollary 4.2 with $\mu \leq 1/2$, let C be the component of $\{|f| < 2\mu\}$ containing c_* , and put $\kappa = \text{cap}(\bar{C})/(2\mu)^{1/n}$. If $\kappa \leq \tau_{k_0}$, where*

$$\tau_k = \frac{\sqrt{2k} - A}{B}, \quad A = \frac{283}{3610}, \quad B = \frac{52029}{9100},$$

then Erdős #1041 holds for f . In particular $\kappa \leq 1/3$ suffices at every arity, and the rational cutoffs $2/5, 12/25, 1/2, 7/12, 16/25, 2/3, 7/10$ suffice at $k_0 = 3, \dots, 9$, rising to $39/40$ at $k_0 = 16$.

Proof. Replace the global input $\text{Area}(K_T) \leq \pi T^{2/n}$ in the proof of Theorem 4.1 by the exact component form $\text{Area}(C) \leq \pi \text{cap}(\bar{C})^2 = \pi \kappa^2 (2\mu)^{2/n}$ of the area–capacity inequality [2, Theorem 6]. Only the high-lift and boundary terms acquire the factor κ ; the low-lift term (CF2) is unchanged. Taking $\lambda = 2$, $r = 1/20$, and the exact bounds above together with $\log 40 < 12641/3402 < (97/50)^2$, gives length $< \sqrt{2/k_0} (A + B\kappa)$. The definition of τ_{k_0} makes the right side at most 2, and for each displayed rational q_k integer arithmetic gives $(A + Bq_k)^2 < 2k$. Discarding $\sqrt{2/k_0} \leq 1$ altogether gives the uniform cutoff $\kappa \leq 1/3$. $\square$

By the component-capacity formula, $\kappa = e^{-\Sigma/n}$ with Σ the sum of exterior Green function values at the roots excluded from C . Indeed, let Ω be the unbounded component of $\mathbb{C} \setminus \bar{C}$, let G be its Green function with pole at infinity, extended by zero off Ω ,

and let $g(\cdot, a)$ be its Green function with pole at $a \in \Omega$. The function $\frac{1}{n} \log \frac{|f|}{2\mu} - G + \frac{1}{n} \sum_a g(\cdot, a)$, summed over the roots $a \in \Omega$ with multiplicity, extends harmonically to Ω , is bounded, and vanishes on $\partial\Omega \subseteq \partial C$, so it vanishes identically; its value at infinity is $\log \kappa + \frac{1}{n} \sum_a G(a)$ by the symmetry $g(\infty, a) = G(a)$. A remaining counterexample with $\mu \leq 1/2$ therefore has $k_0 \leq 16$ and exterior Green defect $\Sigma < n \log(40/39)$ at the top arity, so it must be simultaneously low-arity and nearly round.

The level window $[\mu, \lambda\mu]$ is the only loss in Theorem 4.1 that the mechanism itself demands, and removing it isolates one clean extremal problem. At the critical level itself (CF1) fails, because the area derivative diverges logarithmically at a critical level.

Problem 4.4 (one-lobe perimeter). Let C be a connected component of $\{|f| \leq \sigma\}$ containing exactly one root of f . Determine whether a constant β , independent of f , of its degree n and of σ , satisfies

$$\mathcal{H}^1(\partial C) \leq \beta \sigma^{1/n} = \beta \operatorname{cap}\{|f| \leq \sigma\},$$

and if so determine the least such constant β_* .

For $f = z^2 - d^2$ one has $\mu = d^2$ and $\rho = d$. For $\sigma < d^2$ the component of $\{|f| \leq \sigma\}$ around d contains one root, and as $\sigma \uparrow d^2$ its boundary increases to one loop of Bernoulli's lemniscate $|z^2 - d^2| = d^2$, of length $d \Gamma(1/4)^2 / (2\sqrt{\pi})$ by the substitution $\varphi(w) = d\sqrt{1+w}$ and $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$. At $\sigma = d^2$ the two loops meet at the critical point, and the closed sublevel set is a single component containing both roots. Hence $\beta_* \geq \sqrt{2} \omega = \Gamma(1/4)^2 / (2\sqrt{\pi}) = 3.70814935 \dots$, where ω is the lemniscate constant. On $z^n - r^n$ the corresponding closed form $(2^{1/n}/n) \sqrt{\pi} \Gamma(1/(2n)) / \Gamma(1/(2n) + 1/2)$ decreases from 3.7081 at $n = 2$ to 2 as $n \rightarrow \infty$.

The quadratic loop is exceeded in degree eight. For $p(z) = z^8 - (3/2)z$ and $\sigma = 1$, the component C of $\{|p| \leq 1\}$ containing the origin lies in $\{|z| < 4/5\}$, because $|p(z)| \geq 6/5 - (4/5)^8 > 1$ on $|z| = 4/5$, and it contains a neighbourhood of the closed disc of radius $5/8$, because $|p(z)| \leq (5/8)^8 + 15/16 < 1$ on that disc. The other roots satisfy $|z|^7 = 3/2$, so C contains exactly one root, and $\mathcal{H}^1(\partial C) > 5\pi/4 > (\pi/2)(1 + \sqrt{2}) \geq \sqrt{2} \omega$. Moreover, for fixed $a > 1$ and all sufficiently large N , the one-root component of $\{|z^N - az| \leq 1\}$ contains every disc of radius $r < 1/a$, and a circle of radius between $1/a$ and 1 isolates it by Rouché's theorem; letting $a \downarrow 1$ shows that every admissible β is at least 2π . Problem 4.4 is the one-root per-component analogue of the Erdős–Herzog–Piranian lemniscate-length problem, whose per-root constant is 2, attained asymptotically by $z^n - 1$.

Theorem 4.5 (conditional removal of the level window). *If the bound of Problem 4.4 holds with constant β for every $\sigma \leq \mu$, then two roots of f are joined inside K_μ by a path of length at most*

$$\left(\frac{8x}{(1-x)^2} + \frac{\beta}{\pi} \log \frac{1}{x} + \beta \right) \rho \quad (0 < x < 1).$$

Proof. At level μ exactly, c_* lies on the boundary of two one-root lobes U_a, U_b , so it suffices to bound the intrinsic distance from a to c_* inside $\overline{U_a}$. Lift from a along a ray to ∂U_a and travel along that Jordan curve to c_* , so that $d(a, c_*) \leq \ell_a(\theta) + \frac{1}{2} \mathcal{H}^1(\partial U_a)$. The

mean lift is $(2\pi)^{-1} \int_0^\mu P_a(\sigma) d\sigma / \sigma$ with $P_a(\sigma) = \mathcal{H}^1(\partial U_a(\sigma))$. Bound P_a by the distortion estimate for $\sigma \leq x\mu$ and by the hypothesis for $\sigma \geq x\mu$, using $\text{cr}_a \leq 4\rho$ and $P_a(\sigma) \leq \beta\sigma^{1/n} \leq \beta\rho$. Doubling for the two lobes and adding $\frac{1}{2}\beta\rho$ twice gives the stated bound. $\square$

Evidence and exact boundary. Theorem 4.1 is ordinary mathematics with no Lean-checked part. Its inputs are the capacity identity for polynomial preimages [2, §2], Pólya’s area inequality [2, Theorem 1], Cauchy–Schwarz, the Koebe distortion theorem for univalent maps, and a finite graph exit lemma. The public source of record is [unconditional constant factor bound](#), which carries every rational verification quoted above. Its containment level is 2μ and its constant is $71/10$, so the theorem itself reaches neither half of the target, and the sharp constant 2 remains open. Corollaries 4.2 and 4.3 do reach the target on their stated regions; each rational cutoff is an integer inequality of the form $(A + Bq)^2 < 2k$, checked exactly. The complementary cells, low first-merge arity together with high component capacity, stay open. For Problem 4.4, 120 sampled configurations over degrees 2 through 10 and five families gave a largest ratio $\mathcal{H}^1(\partial U)/(\sqrt{2}\omega\rho)$ of 0.9999997747, at quadratics. The degree-eight example above exceeds the quadratic constant, so the sampled maximum is a local one. The adjacent classical literature on lemniscate length concerns the arclength of the level curve $\{|p| = 1\}$, which is Erdős #114. Fryntov and Nazarov recall its history [15, Introduction]: Dolzhenko’s bound $4\pi n$, Pommerenke’s bound $74n^2$ of 1961, and Borwein’s bound $8\pi en$ [13]. Eremenko and Hayman proved $9.173n$ [14, Theorem 1], Fryntov and Nazarov the asymptotically sharp $2n + o(n)$ [15], and Tao resolved the problem for large n [16]. These results bound the length of a level curve; a dated search on 2 September 2026 located no constant-factor root-pair bound in arbitrary degree.

5 Degree three

Theorem 5.1 (the cubic case). *Let $f(z) = \prod_{j=1}^3 (z - z_j)$ with $|z_j| < 1$, the roots listed with multiplicity. Then two listed root occurrences are joined inside $\{|f| < 1\}$ by a polygonal path of length strictly below 2. If f is squarefree the two are distinct.*

[The complete cubic construction is checked in Lean, including its polygonal path, strict variation bound, and repeated-root case.](#)

Proof. Multiple roots give a constant path, so assume f squarefree. A component containing two roots contains a critical point, so the Erdős–Herzog–Piranian component lemma [4, remark before Problem 5, p. 139] supplies a critical point with critical-value modulus below one.

Suppose first that f' has two distinct zeros. Choose a critical point c minimising $|f(c)|$, write the other one as $c + \delta$, and put $v = f(c)$, so $0 < |v| < 1$. Monicity gives the exact expansion $f(c + d) = d^3 - \frac{3}{2}\delta d^2 + v$. Choose α with $\alpha^3 = v$ and set $b = \delta/\alpha$; dividing by v gives $f(c + \alpha w)/v = P_b(w)$ with

$$P_b(w) = w^3 - \frac{3}{2}bw^2 + 1.$$

At the other critical point $f(c + \delta) = v(1 - b^3/2)$, so minimality of $|v|$ is exactly the hypothesis $|1 - b^3/2| \geq 1$.

Under that hypothesis P_b has at least two zeros, with multiplicity, in the closed unit disc. In the strict region a unit-circle zero w would give $b = \frac{2}{3}(w + w^{-2})$, and with $w^3 = e^{2iu}$ and $c_u = \cos u$ a direct calculation gives $|1 - b^3/2|^2 = 1 + \frac{64}{729}c_u^4(16c_u^2 - 27) \leq 1$, an equality only at $c_u = 0$, where $b = 0$. So the strict region has no unit-circle zero. The radial deformation $b \mapsto sb$, $s \geq 1$, stays in the strict region, since $z = b^3/2$ and $|1 - z|^2 > 1$ give $|1 - s^3z|^2 - 1 = s^3(s^3|z|^2 - 2\Re z) > 0$. For large s , Rouché's theorem on $|w| = 1$ compares P_{sb} with $-\frac{3}{2}sbw^2$, whose modulus $\frac{3}{2}s|b|$ exceeds $2 \geq |w^3 + 1|$, so P_{sb} has exactly two zeros in the open unit disc; no zero crosses the circle along the deformation. The equality case follows by letting s decrease to one and using continuity of the root multiset.

If $P_b(w) = 0$ and $0 \leq t \leq 1$, then $P_b(tw) = 1 - t^2 - t^2(1 - t)w^3$, so $|w| \leq 1$ gives $|P_b(tw)| \leq (1 - t^2) + t^2(1 - t)|w|^3 \leq 1 - t^3 \leq 1$. The whole segment from 0 to w is therefore safe. Selecting two normalized roots w_1, w_2 with $|w_i| \leq 1$, the two segments from c to $c + \alpha w_i$ lie in $\{|f| \leq |v|\} \subset \{|f| < 1\}$ and have combined length at most $2|\alpha| = 2|v|^{1/3} < 2$.

If instead f' has a double zero c , then $f(c + d) = d^3 + v$ and the roots are $c + \alpha$, $c + \alpha\omega$, $c + \alpha\omega^2$. Averaging their squared moduli gives $|c|^2 + |\alpha|^2 < 1$, so $|\alpha| < 1$; any two radial spokes have total length $2|\alpha| < 2$, and along either spoke the value has modulus $|v|(1 - t^3) < 1$ away from the root endpoint. $\square$

Evidence and exact boundary. The cubic reduction, zero-count lemma and path assembly are now checked in `ErdosProblems/Erdos1041/PaperCubicCompletion.lean`, with endpoint `cubic_paper_complete`; the supporting root count is in `PaperCubic-Schur.lean`. The [monic-cubic endpoint](#) also discharges the root enumeration: monicity and degree three supply the factorisation over $\mathbb{C}$, with multiplicities retained. The connector therefore starts from the polynomial and its open-disc root condition alone; squarefreeness gives distinct root endpoints. The earlier research-corpus presentation is historical working material, rather than the present proof boundary. Pendyala proves the degree-four case [11, Thm. 1, p. 1]; whether degree three was already recorded on the Erdős Problem #1041 discussion thread or in [4] has not been adjudicated, so the priority question here is open. The argument uses the sparse normalized form $w^3 - \frac{3}{2}bw^2 + 1$, and a higher-degree proof needs a replacement for that sparsity.

6 An all-degree critical-value separation theorem

The first result is the strongest direct parent-theorem regime in this note. It is not a statement about a sampled family or a fixed degree: the condition is an exact inequality on the other critical values after normalising one simple selected hub.

Theorem 6.1 (disk-family critical-value separation). *Let P be a polynomial of degree $n \geq 3$ whose leading coefficient has modulus one, with*

$$P(0) = 1, \quad P'(0) = 0, \quad P''(0) \neq 0.$$

Fix $w_0 \in [0, 1]$ and $S > \max(w_0, 1 - w_0)$. Suppose every other critical point $d \neq 0$ satisfies

$$|P(d) - w_0| \geq S. \tag{4}$$

Put $p = w_0(1 - w_0)$. The two local solutions of $P(Z(\xi)) = 1 - \xi^2$, $Z(0) = 0$, continue along the real segment to one injective root-to-root connector Γ . Its endpoints are distinct roots, $\Gamma \subseteq \{|P| \leq 1\}$, and

$$\text{length}(\Gamma)^2 \leq 2 \left(\frac{S}{n-1} \right)^{2/n} \log \frac{S^2 + S + p}{S^2 - S + p}. \quad (5)$$

Consequently the connector is shorter than 2 whenever

$$\left(\frac{S}{n-1} \right)^{2/n} \log \frac{S^2 + S + p}{S^2 - S + p} < 2. \quad (6)$$

Proof. Let $Q = D(w_0, S)$ and let U be the component of $P^{-1}(Q)$ containing 0. Both 0 and 1 lie in Q , and (4) says that 0 is the only critical point in U . A component of a polynomial lemniscate is simply connected, by the maximum principle. The proper map $P : U \rightarrow Q$ therefore has degree two: the component-wise Riemann–Hurwitz count in the proof of [8, Proposition 2.1], applied to $(P - w_0)/S$, gives $1 = \deg(P|_U) - 1$ because the simple critical point contributes the only unit of ramification.

Set $a = 1 - w_0$. On U , the function $1 - P$ has only its double zero at 0, so it has a single-valued square root

$$\xi(z)^2 = 1 - P(z).$$

The map ξ is proper and locally injective from U onto

$$\tilde{Q} = \{\xi : |\xi^2 - a| < S\}.$$

Here $\tilde{Q}$ is the connected simply connected double cover of $D(a, S)$ branched at 0. Hence ξ is a conformal bijection; its inverse Z is the continued saddle branch. Since $[-1, 1] \subset \tilde{Q}$, the curve $Z([-1, 1])$ joins the two distinct points over $P = 0$, and $P(Z(\xi)) = 1 - \xi^2 \in [0, 1]$ gives its containment.

The next change of coordinate puts the exact curve, rather than a larger disc, into Bergman normal form. The map

$$M(w) = \frac{Sw}{S^2 + aw - a^2}$$

sends $D(a, S)$ conformally to the unit disc and fixes 0. Choosing the analytic square root gives a conformal bijection

$$\zeta(\xi) = \xi \sqrt{\frac{S}{S^2 + a\xi^2 - a^2}} : \tilde{Q} \rightarrow \mathbb{D}.$$

It sends $[-1, 1]$ to $[-q, q]$, where

$$q^2 = \frac{S}{S^2 + p}.$$

For $\Phi = Z \circ \zeta^{-1} : \mathbb{D} \rightarrow U$, the Bergman segment inequality, proved by the kernel estimate of Section 3 with the double integral taken over $[-q, q]^2$, gives

$$\begin{aligned} \text{length}(\Gamma)^2 &\leq \frac{2}{\pi} \log \frac{1 + q^2}{1 - q^2} \text{Area}(U) \\ &= \frac{2}{\pi} \log \frac{S^2 + S + p}{S^2 - S + p} \text{Area}(U). \end{aligned} \quad (7)$$

It remains to bound the area of this particular component. For $1 - w_0 < S' < S$, let U' be the component of $\{|P - w_0| < S'\}$ containing 0. It is a proper regular component carrying exactly two zeros of $P - w_0$, counted with multiplicity. The exterior-fibre capacity gap for a proper k -zero component, here with $k = 2$, gives

$$\frac{\text{cap}(\overline{U'})^n}{S'} < \frac{k}{2n - k} = \frac{1}{n - 1}.$$

Pólya's area-capacity inequality $\text{Area}(K) \leq \pi \text{cap}(K)^2$ [2, Theorem 6] then yields $\text{Area}(U') < \pi(S'/(n - 1))^{2/n}$. These components exhaust U as $S' \nearrow S$, so

$$\text{Area}(U) \leq \pi \left(\frac{S}{n - 1} \right)^{2/n}.$$

Substitution into (7) proves (5), and (6) makes its right side strictly less than 4. $\square$

Corollary 6.2 (uniform radius $4/3$). *Inequality (6) holds for every $n \geq 3$, every $w_0 \in [0, 1]$, and every $4/3 \leq S \leq 2$. Thus, if f is monic with roots in the open unit disc, c is a simple critical point with $v = f(c) \neq 0$ and $|v| < 1$, and*

$$\left| \frac{f(d)}{v} - w_0 \right| \geq \frac{4}{3}$$

for every other critical point d , then two roots of f are joined inside $\{|f| < 1\}$ by a curve of length strictly below 2.

Proof. For $S \geq 4/3$ and $p \geq 0$,

$$\frac{S^2 + S + p}{S^2 - S + p} \leq \frac{S + 1}{S - 1} \leq 7.$$

Also $S/(n - 1) \leq 1$ when $S \leq 2$ and $n \geq 3$, while $\log 7 < 2$. This proves (6). For the unnormalised polynomial, set $z = c + \rho w$, where $\rho = |v|^{1/n}$; then $P(w) = f(z)/v$ has leading coefficient $|v|/v$, of modulus one. Scaling the connector multiplies its length by $\rho < 1$ and takes $\{|P| \leq 1\}$ to $\{|f| \leq |v|\} \subset \{|f| < 1\}$. $\square$

Taking $S = 2$ gives the earlier branch-centred criterion $|1 - f(d)/v| \geq 2$ in every degree $n \geq 3$. If c minimises the nonzero critical-value modulus, the Fekete-resultant bound gives $|v| \leq R^n < 1$ when all roots lie in a disc of radius $R < 1$; for a general selected hub, the condition $|v| < 1$ must remain explicit. In degree three the branch-centred choice $w_0 = 1$ already works at $S = 6/5$, since

$$\left(\frac{3}{5} \right)^{2/3} \log 11 < 2.$$

This sharper cubic constant is part of the numerical Lean kernel cited below.

The separation hypothesis of Theorem 6.1 constrains critical values only. Roots and critical points can remain uniformly separated while two distinct critical values coalesce. An exact example is

$$g(z) = z^4 - \frac{z^3}{262144} - 2z^2 + \frac{3z}{262144} + \frac{1}{2}, \quad g'(z) = 4(z + 1)(z - 3/1048576)(z - 1).$$

Its four roots lie respectively in $(-3/2, -5/4)$, $(-3/4, -1/2)$, $(1/2, 3/4)$, and $(5/4, 3/2)$. Thus the four roots and three critical points are pairwise more than $1/4$ apart, but the critical values at -1 and 1 differ by only $1/65536 < 3/65536$. Root and critical-point separation alone therefore cannot discharge the value-plane hypothesis of the theorem.

Evidence and exact boundary. The two-sheeted component, its square-root uniformisation, the Bergman segment inequality, the exterior-fibre capacity gap, Pólya’s area–capacity inequality and the exhaustion in the proof are ordinary mathematics. Their complete source proof is the immutable [disk-family critical-value separation record](#). The separate [Lean numerical kernel](#) checks the coefficient bound for $n \geq 3$, $4/3 \leq S \leq 2$ and $p \geq 0$, and the implication from the squared length bound to length below 2. It does not formalise the analytic hypotheses producing (5). The adjacent axiom-audit source names those numerical declarations; it is not an axiom audit of the ordinary analytic theorem.

The method stops when no disk centred on $[0, 1]$ isolates the selected critical value, and it assumes the selected saddle is simple. It therefore leaves the near-tie and multiple-saddle strata, makes no sharpness claim for the uniform threshold $S = 4/3$, asserts no existence of a separated disk, and does not prove an unrestricted admissible-hub selector, a COVER theorem, or Erdős #1041.

The literature disposition for Dubinin’s Theorem 1 is settled here. The source is [12], Theorem 1 on printed page 85 of the POMI original. It concerns a holomorphic function that gives a full n -fold covering of an annulus $t_1 < |w| < t_2$, and, in the notation of that theorem, with E its explicitly defined complementary set, it states

$$\left(\frac{t_2}{t_1}\right)^{2/n} \leq \frac{m(E \cup D)}{m(E)}.$$

The full covering and the explicit complementary set are hypotheses of the theorem. Tao cited it on the Erdős Problem #1041 discussion page on 25 March 2026 for the relative scaling factor $s^{2/n}$ between the areas of two nested sublevel sets, which is the reading of the displayed inequality in which E and $E \cup D$ are those two sublevel sets and the covering hypothesis holds. No step of this note uses that relative inequality. The disk-family theorem instead uses Pólya’s absolute area–capacity inequality on one proper component, after the exterior-fibre argument supplies its strict capacity gap. The absolute sublevel inequality $\text{Area}\{|P| < T\} \leq \pi T^{2/n}$ remains the area input to Theorem 3.1 and to Theorem 4.1. Dubinin’s theorem is a neighbouring result under a full covering hypothesis and supplies no root connector.

7 Three exact solved polynomial families

The next three theorems are unconditional subcases of the path problem, not proxies for the unrestricted conjecture. They are ordered by mathematical signal: first an all-degree sharp theorem with equality configurations, then a complete degree-five sparse family, and finally an infinite tower of translated quotient fibres. The date of their formalization has no bearing on that order. In each case a displayed finite proposition is the exact Comparator-selected Lean endpoint. The ensuing normalization and path theorem is ordinary mathematics and is stated separately.

7.1 Sharp collinear zeros

Put

$$r_n = \cos \frac{\pi}{2n}, \quad C_n = \frac{1}{2^{n-1}r_n^n}.$$

Theorem 7.1 (checked Chebyshev endpoint). *Let $m \geq 0$, let $p \in \mathbb{R}[X]$ be monic of degree $m + 2$, and let*

$$-1 < c_0 < \dots < c_m < 1, \quad |c_i| \leq 1.$$

Suppose $p(-1) = p(1) = 0$ and $p(c_i)p(c_{i+1}) < 0$ for $0 \leq i < m$. Then

$$\min_{0 \leq i \leq m} |p(c_i)| \leq C_{m+2}.$$

This is exactly the statement selected as [the formal Chebyshev endpoint](#). It is the kernel-facing alternation endpoint, not yet the geometric theorem.

Theorem 7.2 (sharp collinear diameter theorem). *Let f be a monic polynomial of degree $n \geq 2$ whose zero occurrences are collinear, and let D be their diameter. Some two adjacent zero occurrences are joined by a segment of length at most D on which*

$$|f(z)| \leq \frac{(D/2)^n}{2^{n-1} \cos^n(\pi/(2n))}. \quad (9)$$

The constant in (9) is best possible in every degree. Equality is attained by affine images of the zeros of T_n whose extreme zeros have distance D .

Proof. A repeated zero gives the constant path, so assume the zero values are distinct. A rigid motion sends their line to the real axis, their midpoint to the origin, and their extremes to $\pm D/2$. With $R = D/2$, the normalized polynomial

$$q(w) = R^{-n} e^{-in\theta} f(m + Re^{i\theta}w)$$

is monic with real zeros $-1 = y_1 < \dots < y_n = 1$, and $|f(m + Re^{i\theta}w)| = R^n |q(w)|$.

Compare q with the monic endpoint-normalized Chebyshev polynomial

$$q_*(x) = \frac{T_n(r_n x)}{2^{n-1}r_n^n}.$$

Both q and q_* vanish at ± 1 , and $|q_*| \leq C_n$ on $[-1, 1]$. For each gap $[y_i, y_{i+1}]$, choose c_i at which $|q|$ is maximal. The signs of $q(c_i)$ alternate. If every gap maximum were larger than C_n , then $q - q_*$ would have the same alternating signs as q at the c_i . It would therefore have a zero between each consecutive pair c_i, c_{i+1} , as well as the two zeros ± 1 . These are n distinct zeros, whereas the leading terms cancel and $\deg(q - q_*) \leq n - 1$; moreover $q - q_*$ is nonzero at every c_i . This contradiction selects a gap on which $|q| \leq C_n$. Scaling back proves (9).

For sharpness, take

$$y_k = \frac{\cos((2k-1)\pi/(2n))}{r_n}, \quad 1 \leq k \leq n.$$

These are the zeros of q_* , have extremes ± 1 , and every adjacent gap contains a scaled Chebyshev extremum where $|q_*| = C_n$. No smaller universal constant can work. $\square$

Corollary 7.3 (collinear Erdős case). *If the zero occurrences of a monic polynomial of degree $n \geq 2$ lie on one line in the open unit disc, two of them are joined by a curve of length strictly below 2 inside $\{|f| < 1\}$.*

Proof. Their diameter satisfies $D < 2$. Also $C_n \leq 1$ for $n \geq 2$, with equality only at $n = 2$, so (9) is strictly below one. The selected segment has length at most $D < 2$. $\square$

Corollary 7.3 already follows from a theorem of Erdős, Herzog and Piranian [4, Theorem 1, p. 126]: if the zeros of a monic polynomial of degree n are real, lie in $[-1, 1]$ and have centroid in $[0, 1]$, then $\{|f| < 1\} \cap \mathbb{R}$ contains an interval holding at least $n/2$ of the zeros. Apply it to the normalized polynomial q , after the reflection $w \mapsto -w$ if the centroid is negative. For $n \geq 3$ two consecutive zeros in that interval are joined by a segment of length at most D inside $\{|f| < (D/2)^n\}$, and for $n = 2$ the segment between the zeros lies in $\{|f| \leq (D/2)^2\}$. Theorem 7.2 adds the sharp level and its equality configurations.

Critical sequences and the extremal configuration. For distinct real zeros, each gap between consecutive zeros contains exactly one zero of q' , and $|q|$ attains its maximum over the gap only there. The points c_i chosen in the proof of Theorem 7.2 are therefore the critical points of q , and the gap maxima are the moduli of its critical sequence $(q(c_1), \dots, q(c_{n-1}))$. A real polynomial with real zeros is determined by its critical sequence up to an increasing real affine change of variable [17, Theorem 1]; the proof uses only the alternation count. In this language, (9) says that a real polynomial of degree n with n distinct real zeros, leading coefficient a and zero diameter D has a critical value of modulus at most $C_n |a| (D/2)^n$.

The comparison polynomial q_* is the monic polynomial of least deviation from zero on $[-1/r_n, 1/r_n]$, characterised by the equioscillation conditions recalled in [17, §1]. All $n - 1$ of its critical values have modulus C_n , so every critical point of q_* lies on the level curve $\{|q_*| = C_n\}$. Extremals with this property also occur in the level-curve length problem, where some extremal polynomial has all its critical points on $\{|p| = 1\}$ [14, Lemma 5], and in the sharp bound for $|f'|$ on a connected sublevel set $\{|f| \leq 1\}$ of a monic polynomial, proved by Eremenko and Lempert [18, Theorem 1], whose equality cases are $e^{-in\theta} T_n(2^{1/n-1} e^{i\theta} z + b)$ with θ real [19, Theorem A].

Formal boundary. Lean checks sign preservation, the alternating root-count mechanism, the scaled Chebyshev endpoint and uniform bound, and Theorem 7.1. It does not check the rigid normalization, the existence of the gap maxima, transport back to the root line, or the equality-node locations used in Theorem 7.2. These are ordinary steps. The sharpness assertion concerns the maximum of $|f|$ on the selected adjacent-root segment, and the displayed configurations attain it; no claim is made that they exhaust the equality cases, and no claim is made that the selected segment is a shortest path in $\{|f| < 1\}$.

7.2 The primitive sparse quintic

The next selector is finite but unusually rigid. For $0 < r < 2$ and $0 \leq s_i \leq 1$, $x_i^2 \leq s_i$, put

$$E_i = s_i^4 (s_i + r^2 + 2rx_i).$$

Theorem 7.4 (checked two-tail energy selector). *Suppose*

$$\sum_{i=0}^4 x_i = -r, \quad \sum_{i=0}^4 (2x_i^2 - s_i) = r^2, \quad \sum_{i=0}^4 (4x_i^3 - 3s_i x_i) = -r^3. \quad (10)$$

Then at least one of the ten pairs $0 \leq i < j \leq 4$ satisfies $E_i < 1$ and $E_j < 1$.

This is the exact finite conclusion selected as [the formal two-tail selector](#). It selects two distinct indices. It does not assert that the corresponding complex root values are distinct.

Theorem 7.5 (primitive sparse quintic). *Let*

$$p(z) = z^5 + az^4 + bz + c$$

and suppose its five zero occurrences $w_0, \dots, w_4$ lie in the closed unit disc. At least two distinct indices satisfy

$$|bw_i + c| \leq 1. \quad (11)$$

If $a \neq 0$, two indices can be chosen with strict inequalities. If $a = 0$, every index satisfies (11), and equality holds exactly when $|w_i| = 1$.

For open-disc zeros, two zero occurrences are joined inside $\{|p| < 1\}$ by a curve of length below 2: use the two radial spokes through 0 when their values are distinct, and the constant path when the selected occurrences have the same value.

Proof. Rotate so that $a = re^{i\phi}$ becomes $r \geq 0$ and write $z_i = e^{-i\phi} w_i$. The missing z^3 and z^2 coefficients and Newton's identities give

$$\sum z_i = -r, \quad \sum z_i^2 = r^2, \quad \sum z_i^3 = -r^3. \quad (12)$$

At a zero,

$$|bw_i + c| = |w_i|^4 |w_i + a|.$$

Thus, for $x_i = \Re z_i$ and $s_i = |z_i|^2$, the squared tail is exactly E_i and (12) becomes (10). If $r > 0$, the third moment also gives $r^3 = |\sum_i z_i^3| \leq 5 < 8$, so $r < 2$ as required by the selector.

For completeness, the selector is driven by the harmonic extension

$$H_r(x, s) = (-r/2 - x)(1 - x)^2 + (1 - s) \left(1 - \frac{r}{4} - \frac{3x}{4} \right).$$

The three moments give the exact sum

$$\sum_{i=0}^4 H_r(x_i, s_i) = 5 - 2r, \quad (13)$$

while $H_r \leq 4 - 2r$ throughout the unit disc. If $E_r(x, s) \geq 1$, then

$$5(1 - s) \leq 2r(x + r/2).$$

Put $d = x + r/2$, $u = 1 - x$, $A = 1 - r/4 - 3x/4$, and $P = -u^2 + (2r/5)A$. The unsafe inequality gives $0 \leq d < 2$. Since $H_r = -du^2 + (1 - s)A$, it is nonpositive when $A \leq 0$; when $A > 0$, the preceding bound gives $H_r \leq dP$. The remaining estimate is the exact sum-of-squares identity

$$\frac{1}{31} - P = \frac{2}{5} \left(d - \frac{38}{31} + \frac{1}{4} \left(u - \frac{3}{31} \right) \right)^2 + \frac{31}{40} \left(u - \frac{3}{31} \right)^2 \geq 0,$$

which gives $P \leq 1/31$, and hence $H_r(x, s) \leq 2/31$ whether P is positive or nonpositive. Thus every unsafe tail has this smaller score. If four tails were unsafe, (13) would give

$$5 - 2r \leq 4 - 2r + \frac{8}{31} < 5 - 2r,$$

a contradiction. This proves the two-index conclusion. The case $r = 0$ is the direct identity $|bw_i + c| = |w_i|^5$.

Finally, at a zero w ,

$$p(tw) = (1 - t)c + (t - t^4)(bw + c) - t^4(1 - t)w^5.$$

For $0 \leq t \leq 1$ the three nonnegative coefficients sum to $1 - t^5$. The tail bound, $|w| \leq 1$, and $|c| \leq 1$ therefore give the radial unit-sublevel path. Under the open-disc hypothesis the relevant bounds and the total spoke length $|w_i| + |w_j|$ are strict. $\square$

Formal boundary. Lean checks the five-index moment hypotheses, the harmonic cap, the sum-of-squares unsafe estimate, and the ten-pair conclusion in Theorem 7.4. The rotation, derivation of Newton moments from the roots, root-tail identity, Abel decomposition, path assembly, and any passage from distinct indices to distinct complex values remain ordinary mathematics.

7.3 Translated cubic quotient fibres

Theorem 7.6 (checked cubic safe spoke). *If $r, s, v \in \mathbb{C}$ have modulus below one, at least one $u \in \{r, s, v\}$ satisfies*

$$|(tu - r)(tu - s)(tu - v)| \leq 1 \quad (0 \leq t \leq 1).$$

This is exactly [the formal cubic safe-spoke theorem](#).

Theorem 7.7 (translated cubic quotient fibres). *Let $q \geq 2$, $h \in \mathbb{C}$, P be monic cubic, and*

$$f(z) = P((z - h)^q).$$

If every zero of f lies in the open unit disc and f has at least two distinct zero values, then two zeros are joined through h by a two-segment path of length below 2 inside $\{|f| < 1\}$. Equivalently this closes the coefficient family

$$(z - h)^{3q} + A(z - h)^{2q} + B(z - h)^q + C$$

in every degree $3q \geq 6$.

Proof. Fix a q th root y of a quotient root and a primitive q th root of unity ζ . The full fibre consists of $h + y\zeta^k$. Since every fibre point lies in the open unit disc,

$$\frac{1}{q} \sum_{k=0}^{q-1} |h + y\zeta^k|^2 = |h|^2 + |y|^2 < 1. \quad (14)$$

Thus $|y| < 1$, and the corresponding quotient root has modulus $|y|^q < 1$. This verifies the open-disc hypothesis for all quotient roots before applying the following selector. The identity retains more information than these separate bounds: $|h| < 1$ and every fibre radius is strictly below $\sqrt{1 - |h|^2}$. Consequently, any two safe fibre spokes constructed below have total length strictly less than $2\sqrt{1 - |h|^2}$. The position of the symmetry centre therefore controls the metric budget; the bound 2 discards this information.

Write $P(w) = (w - r)(w - s)(w - v)$ and assign to r the charge $A_r = \Re(\overline{rs + v})$, cyclically. The exact sum is

$$A_r + A_s + A_v = |r + s + v|^2 - (|r|^2 + |s|^2 + |v|^2) > -3,$$

so one charge, say A_r , exceeds -1 . For $0 \leq t \leq 1$, AM–GM and the distance-square identity give

$$|tr - s||tr - v| < 1 + t + t^2.$$

Consequently, for $0 \leq t < 1$,

$$|P(tr)| < (1 - t)(1 + t + t^2) = 1 - t^3 \leq 1. \quad (15)$$

At $t = 1$ the polynomial vanishes. The entire spoke is therefore strictly contained, including its origin endpoint.

For a nonzero safe quotient root from (15), two distinct fibre points satisfy

$$f(h + ty\zeta^k) = P(t^q r),$$

and their two spokes through h have total length $2|y| < 2\sqrt{1 - |h|^2} \leq 2$. If the selected quotient root is zero, choose a nonzero quotient root and use the direct estimate $|P(ts)| < 2t(1 - t) \leq 1/2$ for $0 < t < 1$, with zero values at both endpoints; if none exists, f has only one distinct zero value, contrary to the hypothesis. $\square$

Formal boundary. Lean checks the charge identity, safe-spoke selection and quartic falsifier, and also [the complete translated cubic fibre theorem](#). The latter supplies the finite fibre construction, root-of-unity average, nonzero-root selection, pullback and explicit rectifiable hub path under the hypotheses displayed above. Its exact source occurs in the successfully compiled dependency closure of [the source-bound Lean build and axiom-audit record](#). The theorem gives length below 2; the sharper $2\sqrt{1 - |h|^2}$ reading in the proof is recorded here as an ordinary consequence of the mean-square identity, not as a separately audited endpoint.

Shared assurance boundary. The three Comparator declarations route respectively to Theorems 7.1, 7.4, and 7.6. They support, but are strictly weaker than, the ordinary assembled Theorems 7.2, 7.5, and 7.7. The displayed Lean endpoints and the assembled geometric theorems therefore have different formalization boundaries.

8 The current frontier: what survives near Fekete configurations

The August 29 public research update changes the useful first reading of this problem. It does not add a proof of Erdős #1041; it identifies the geometric regime in which the remaining difficulty is concentrated and kills several attractive but false shortcuts. The source of the update is pinned to commit [f214a6b4](#) and the dated synthesis is [the #1041 frontier note](#). The distinctions below are part of the result: a theorem, a refutation, a certificate, and a measurement do not have interchangeable force.

Near-Fekete stability removes the radial distraction

Write $a_i = \rho_i u_i$, with $|u_i| = 1$, and put $D = |\text{disc}(f)|/n^n$. The quantitative Fekete–Hadamard estimate in [near fekete radial angular split](#) says that, when $D \geq 1 - \eta$ and $\eta \leq 1/(80n^2)$,

$$1 - \rho_i^2 \leq \frac{n\eta}{n-1}, \quad |a_i - a_j| \geq \frac{2 - 2\sqrt{\eta} - n\eta}{n-1},$$

and the roots lie within $7\sqrt{\eta}$ of a rotated regular n -gon, after a bijection. The mechanism is a normalised Vandermonde Gram determinant: near equality in Hadamard’s inequality forces both the radii and the angular separation to be near extremal. This is an ordinary mathematical result in the public research corpus, not a new Lean declaration.

The associated radial monotonicity theorem is the more important reader conclusion. Under the stronger near-Fekete condition $\eta \leq 1/(10n^4)$ used by this corollary, replacing the radii by one can only increase the values along the corresponding radial spoke. Thus radial deficits help containment; the unresolved near-Fekete problem is angular. On exact regular-gon directions this becomes unconditional for $n \leq 6$: arbitrary radii in the stated band give

$$|f(s \omega \zeta^i)| \leq 1 - s^n \quad (0 \leq s \leq \rho_i),$$

so every two-radii path is contained. The good spoke in the general on-circle averaging identity may move with s , however; that is why an existence statement at one level does not supply a fixed pair of roots.

A bounded-radius concyclic class

There is nevertheless a clean theorem for a different on-circle mechanism. Let f be monic of degree $n \geq 3$ with distinct zeros on a circle of radius ρ . If $2\rho^n \leq 1$, two adjacent zeros are joined by their straight chord, whose length is at most $2\rho \sin(\pi/n) < 2$, and the chord lies in $\{|f| < 1\}$. (If a zero is repeated, the short-connection conclusion is immediate; the distinct case is the substantive one.)

After normalisation to the unit circle, self-inversive realification and alternation against $z^n - c$ select a zero-free gap on which the polynomial modulus is at most 2; a harmonic normal-derivative argument transfers this arc bound strictly to the chord. The complete ordinary proof is [concyclic alternation](#).

The argument is an ordinary proof outside Lean. The [exact-rational checker](#) checks finitely many load-bearing identities and configurations, while the [numerical checker](#)

is regression and stress-test evidence. The arc constant 2 is sharp on the regular n -gon, so this chord argument does not reach radii tending to 1; the unrestricted concyclic case and Erdős #1041 remain open.

Cyclic trinomial fibres

The spoke identity and strict bound are formalized in [trinomial root spoke factorization](#) and [trinomial root spoke norm lt one of norm lt one](#).

There is also a positive structured family in every cyclic lift degree. Fix integers $1 \leq r \leq m$ and $q \geq 1$, and consider the translated polynomial

$$f(z) = (z - h)^{qm} + a(z - h)^{qr} + c.$$

Writing $w = (z - h)^q$ reduces the root equation to $w^m + aw^r + c = 0$. At such a quotient root the middle coefficient can be eliminated exactly: for $0 \leq u \leq 1$,

$$u^m w^m + au^r w^r + c = (1 - u^r)c - (u^r - u^m)w^m.$$

The two coefficients on the right are nonnegative and have total at most one. The [formal strict-spoke theorem](#) therefore proves that, if $|w| < 1$ and $|c| < 1$, the complete radial spoke is strictly contained in the open unit lemniscate. This is an all-root statement; no Vieta-small root selection is needed for the containment bound.

For a nontrivial cyclic fibre, two selected displacements y_1, y_2 with $|y_1|, |y_2| < 1$ also satisfy $|y_1| + |y_2| < 2$. The formal source checks the factorization, strict spoke estimate, and this metric budget. It does not yet check the ordinary finite argument selecting a suitable quotient root and two distinct members of its fibre, lifting both quotient spokes, or assembling their union as a path in the relevant component. Thus the Lean theorem is the load-bearing analytic kernel for this translated cyclic-trinomial family, not the complete path theorem and not a proof of unrestricted Erdős #1041.

Coefficient-controlled cyclic tetranomial spokes

The first additional lower monomial admits an exact Abel-tail criterion. Let

$$g(w) = w^m + aw^r + bw^s + c, \quad m > r > s \geq 1,$$

assume all roots of g lie in the open unit disk, and let w_1, w_2 be roots of the two smallest moduli. If

$$|c| + |b||w_2|^s < 1,$$

then the complete radial spokes from w_1 and w_2 to the origin lie in $\{|g| < 1\}$, and their broken line has length strictly below 2. The simpler coefficient condition

$$|b| + |c| \leq 1$$

makes every root spoke safe; the coefficient a is unrestricted.

The mechanism is visible in one identity. At a root w and for $0 \leq u \leq 1$,

$$\begin{aligned} g(uw) &= (1 - u^s)c - (u^s - u^r)(aw^r + w^m) - (u^r - u^m)w^m, \\ aw^r + w^m &= -(c + bw^s). \end{aligned}$$

The three scalar coefficients are nonnegative and sum to $1 - u^m$. Thus the root-dependent budget $|c| + |b||w|^s < 1$, together with $|w| < 1$ and $|c| < 1$, puts every vector in the Abel decomposition strictly inside the unit ball. Lean checks the [factorization](#) and the resulting [strict spoke theorem](#) under the exact weak exponent hypotheses $1 \leq s \leq r \leq m$; it also checks the coefficient-only corollary above.

There is a stronger, genuinely two-index formal theorem. Let S index a finite family of roots, $N = |S| \geq 2$, and

$$M = \sum_{i \in S} w_i^s.$$

The [formal energy identity](#) gives

$$\sum_{i \in S} |c + bw_i^s|^2 = N|c|^2 + |b|^2 \sum_{i \in S} |w_i^s|^2 + 2 \operatorname{Re}(\bar{c} b M).$$

Consequently, if every $|w_i| < 1$, $|c| < 1$, and

$$N(|b|^2 + |c|^2) + 2 \operatorname{Re}(\bar{c} b M) < N - 1,$$

then two distinct indices $i, j \in S$ satisfy $|c + bw_i^s| < 1$ and $|c + bw_j^s| < 1$. Feeding those inequalities into the [formal Abel-spoke consumer](#) proves that both complete root spokes lie strictly in $\{|g| < 1\}$. The signed cross term is essential: this interface can certify configurations outside the coefficient-only triangle $|b| + |c| \leq 1$. The formal hypotheses do not require $i \mapsto w_i$ to be injective, so distinct indices need not denote distinct root values; the ordinary two-root consequence requires a root enumeration without repetition.

For $q \geq 2$ and $f(z) = g((z - h)^q)$, the ordinary regular-fibre mean-square identity puts every quotient root inside the unit disk. Lifting two selected quotient spokes gives a path through h whose two displacement lengths sum to less than 2. The new Lean theorem assumes the finite root family, its signed moment M , and the displayed budget; it does not derive that budget from arbitrary tetranomial coefficients, prove that the supplied family is a complete root multiset, perform cyclic-fibre lifting, or construct the final path object. Without the signed-moment, root-dependent, or coefficient-only budget, the tetranomial case remains open; this family does not solve unrestricted Erdős #1041.

Translated quartic quotient fibres

Pendyala's degree-four theorem [11, Theorem 1], with his four-point radial lemma [11, Lemma 1], can also be lifted through every nontrivial cyclic power. Let P be a monic quartic, let $q \geq 2$, and set

$$f(z) = P((z - h)^q).$$

If all listed zeros of f lie in the open unit disk, averaging the squared moduli over each complete q -point fibre shows that every quotient root w of P satisfies $|w| < 1$. Pendyala's chord-or-radial proof then supplies the quotient geometry, in two cases. If two quotient roots satisfy $|w_i - w_j| < 1$, every point w of the chord $[w_i, w_j]$ lies in the unit disc; the two endpoint factors of $|P(w)|$ have product at most $|w_i - w_j|^2/4$ and the other two factors are below 2, so $|P(w)| \leq |w_i - w_j|^2 < 1$. Otherwise the quotient roots are pairwise at distance at least 1. Choosing R with $\max_k |w_k| < R < 1$ and applying the four-point

radial lemma to the points w_k/R gives distinct i, j with $|P(tw_i)| \leq R^4$ and $|P(tw_j)| \leq R^4$ for $0 \leq t \leq 1$, so the radial segments $[0, w_i]$ and $[0, w_j]$ lie in $\{|P| < 1\}$. The extra issue is metric: a short chord in the w -plane need not lift isometrically through $y \mapsto y^q$.

Put $\alpha = 1/q$. Along a quotient chord whose supporting line has distance d from the origin, the root-lift density is controlled pointwise by

$$(\sqrt{d^2 + x^2})^{\alpha-1} \leq x^{\alpha-1} \quad (x > 0),$$

and Lean checks the [exact primitive](#)

$$\alpha \int_0^A x^{\alpha-1} dx = A^\alpha.$$

Splitting at the perpendicular foot, and at the origin when the chord crosses it, yields the ordinary power-map chord-lift estimate

$$\text{length}(\widetilde{[a, b]}) \leq |a|^{1/q} + |b|^{1/q}.$$

The [formal endpoint consumer](#) also proves that if $0 \leq a, b < 1$, $\alpha > 0$, and a candidate length L is at most $a^\alpha + b^\alpha$, then $L < 2$. The two branches are handled separately, and the distinction matters. The chord-lift estimate applies to a single straight quotient chord, and it bounds the lifted curve by the sum of the two endpoint radii. It does not bound the lift of an arbitrary broken line with a nonzero interior vertex by that sum, so the radial branch is kept in its own form: the fibre bound puts every quotient root inside a centred disc of radius strictly below one, and Pendyala's four-point radial construction is applied there, with the lifted spokes measured through the origin of the quotient plane. With the two branches separated in that way both survive the power-map lift, giving the asserted short path for every translated quartic quotient fibre, in each degree $4q \geq 8$.

The attribution and proof boundary are exact. Pendyala proves the quartic geometric theorem and its four-point radial lemma. The local Lean module checks the antitone density inequality, its integral, the strict powered endpoint budget, and the final length fan-in. It does not formalize Pendyala's geometric lemma, the continuous covering-space construction of the root lift, or the ordinary chord/radial case assembly. This is a structured all-scale family, not a proof of unrestricted Erdős #1041.

A critical-value budget in every degree

The critical values satisfy a sharp aggregate bound in every degree $n \geq 2$. This gives a budget for the merge levels; it does not bound the lengths of the paths that reach them.

Theorem 8.1 (critical-value budget in every degree). *Let f be monic of degree $n \geq 2$, with roots in a closed disc of radius $R \geq 0$. If $c_1, \dots, c_{n-1}$ are its critical points counted with multiplicity, then*

$$\sum_{j=1}^{n-1} |f(c_j)|^{2/(n-1)} \leq (n-1)R^{2n/(n-1)}. \quad (7)$$

Consequently the lower exponents used elsewhere in the record satisfy

$$\sum_{j=1}^{n-1} |f(c_j)|^{1/(n-1)} \leq (n-1)R^{n/(n-1)} \quad (8)$$

and

$$\sum_{j=1}^{n-1} |f(c_j)|^{1/n} \leq (n-1)R.$$

The constant is attained by $f(z) = (z - \tau)^n - \lambda$ with enclosing disk centred at τ and radius $R = |\lambda|^{1/n}$.

The finite inequality behind the theorem concerns arbitrary points of the disk, without asking them to arise as critical points. Its quadratic form is

$$\sum_{j=1}^m \left(\prod_{k=1}^m |1 - \bar{c}_j c_k| \right)^{2/m} \leq m, \quad |c_j| \leq 1.$$

The following pointwise bound explains why these finite inequalities control critical values in every degree.

Lemma 8.2 (reflected-derivative bound). *If f is monic of degree $n \geq 2$ with roots in the closed unit disk, and $c_1, \dots, c_{n-1}$ list its critical points with multiplicity, then*

$$|f(c_j)| \leq \prod_{k=1}^{n-1} |1 - \bar{c}_j c_k|. \quad (9)$$

The reflected-derivative inequality is checked in Lean, including closed-disc roots and critical-point multiplicities.

Proof. First put all roots a_i strictly inside the disk. Gauss–Lucas does the same for the c_k , so

$$N(z) = nf(z) - zf'(z), \quad G(z) = n \prod_k (1 - \bar{c}_k z)$$

have $G \neq 0$ on the closed disk. On $|\zeta| = 1$, $|G(\zeta)| = |f'(\zeta)|$, while

$$\operatorname{Re} \frac{\zeta f'(\zeta)}{f(\zeta)} = \sum_i \left(\frac{1}{2} + \frac{1 - |a_i|^2}{2|1 - a_i \bar{\zeta}|^2} \right) \geq \frac{n}{2}.$$

Writing the logarithmic derivative as w , the identity $|n - w|^2 - |w|^2 = n^2 - 2n \operatorname{Re} w \leq 0$ gives $|N| \leq |G|$ on the circle. The maximum-modulus principle applied to N/G gives the same comparison inside. At c_j , $N(c_j) = nf(c_j)$; dividing by n and conjugating each factor proves (9). For closed-disc roots apply this result to $f_r(z) = r^n f(z/r)$, $0 < r < 1$, whose critical points are rc_k :

$$r^n |f(c_j)| \leq \prod_k |1 - r^2 \bar{c}_j c_k|.$$

Let $r \uparrow 1$. This also handles boundary critical points and all multiplicities without selecting local branches. $\square$

Set $m = n - 1$. The weighted Poisson proof below gives the quadratic finite inequality for every $m \geq 1$. Gauss–Lucas and Lemma 8.2 then give $\sum_j |f(c_j)|^{2/m} \leq m$ on the unit disc. Applying this to $R^{-n}f(h + Rz)$ and scaling back proves (7) for $R > 0$; if $R = 0$, all critical values vanish.

The two displayed consequences are the lower power means of the same nonnegative m -tuple. Equivalently, after unit-disc normalization, apply the monotonicity of normalized L^p means from exponent $2/m$ first to $1/m$ and then to $1/(m + 1) = 1/n$. Scaling back contributes respectively $R^{n/m}$ and R . Thus the older exponent is retained as a consequence, rather than as the principal theorem.

The quadratic mean and both lower-exponent consequences are checked as [the complete critical-value mean](#). This includes arbitrary centre, zero radius, and critical points counted with multiplicity. The exact source hash and the directly audited target alias match [the source-bound Lean build and axiom-audit record](#); the source header’s earlier candidate status predates that successful run.

The stronger exponent controls concentration of the critical values. For $R > 0$, write $r_j = |f(c_j)|^{1/n}$ and $p = 2n/(n - 1)$. Then

$$\sum_j (r_j/R)^p \leq n - 1, \quad \#\{j : r_j \geq tR\} \leq \frac{n - 1}{t^p} \quad (t > 0).$$

The counting bound follows by retaining just those summands. Thus a large critical value consumes more of the budget than the linear estimate records; the radial equality family has every $r_j = R$. This distributional control still leaves the geometry of the joining paths to be supplied.

The weighted Poisson proof and the small-cardinality conclusions. Let $w_j \geq 0$, $\sum_j w_j = 1$, and $|c_j| \leq 1$, with repetitions allowed. Write $G(z) = \prod_k |1 - \overline{c_k}z|^{w_k}$, taking every zero exponent factor to be one. For strictly interior centres choose analytic logarithms zero at the origin and put

$$g(z) = \exp \sum_j w_j \log(1 - \overline{c_j}z) = 1 + \sum_{\nu \geq 1} a_\nu z^\nu, \quad P = \sum_j w_j P_{c_j}.$$

On the unit circle $P = 1 - 2\Re(\zeta g'/g)$. Weighted Poisson majorisation, followed by absolutely justified circle integration, gives

$$\sum_j w_j G(c_j)^2 \leq \int |g|^2 P \, dm = 1 - \sum_{\nu \geq 1} (2\nu - 1) |a_\nu|^2.$$

All functions are analytic on a neighbourhood of the closed disc in this interior case. For closed-disc centres replace c_j by rc_j , $0 < r < 1$. The actual coefficient of degree ν becomes $r^\nu a_\nu$, where the fixed coefficients are defined from the analytic function near zero. Pass to $r \uparrow 1$ first with an arbitrary finite coefficient sum. Nonnegative finite deficits then give summability and the infinite inequality; no boundary holomorphic logarithm is required. Lean checks the [exact radial transport of each Cauchy coefficient](#), the [finite boundary-deficit inequality](#), and the [summable full boundary deficit](#).

If equality holds, all positive-degree coefficients vanish, so $g = 1$ near zero and $\sum_j w_j \overline{c_j}/(1 - \overline{c_j}z) = 0$ there. Group equal centres before clearing denominators. A

nonzero centre of positive total weight gives a nonzero pole coefficient, which is impossible. Thus equality is equivalent to $c_j = 0$ on positive support. In particular, strictly positive weights force all centres zero; $(w, c) = ((1, 0), (0, 1))$ shows why this wording cannot be extended to zero weights. The exact grouped-rigidity source is preserved, not replaced by an argument requiring distinct original centres.

For $M = \sum_j w_j G(c_j)$ the identity

$$M^2 + \sum_j w_j (G(c_j) - M)^2 = \sum_j w_j G(c_j)^2$$

gives weighted Cauchy–Schwarz, with equality precisely when the observations are constant on positive support. Equal weights therefore give the quadratic finite inequality above for every $m \geq 1$, with equality only at the zero configuration. The three-point theorem¹ and four-point theorem state these conclusions with their equality cases.

The previous three-point Hölder proof is withdrawn as an alternative proof from this record. Its scalar defect identity survives independently, but that algebra alone is not credited with the omitted Hölder step or the whole alternative argument. The four-point matching and numerical split are likewise not used. The present argument replaces those passages; it does not claim to formalise their mechanisms.

Poisson majorization and termwise integration. For g holomorphic on the unit disc and continuous on its closure, put $P_c(\zeta) = (1 - |c|^2)/|\zeta - c|^2$ for $|\zeta| = 1$. If $|c| < 1$,

$$|g(c)|^2 \leq \frac{1}{2\pi} \int_0^{2\pi} P_c(e^{it}) |g(e^{it})|^2 dt.$$

The [norm-square majorization](#) uses the Poisson formula for a real part and the non-negativity of a square. The [weighted kernel identity](#) identifies the pointwise Poisson mixture. Separately, [absolute series transport](#) justifies termwise circle integration when the terms are circle integrable and admit a summable uniform norm bound; its Taylor double-product form assumes absolute summability of both coefficient sequences. The checked analytic assembly is accompanied by an 80-declaration audit. The repaired R10 dependency chain and the R11 aggregate and smoke target pass. Across the R10, R11, and closure audits, 320 named prints report only `propext`, `Classical.choice`, and `Quot.sound`.

A uniform central region in every degree. For every $m \geq 1$, the [central-region theorem with its analytic inputs discharged](#) proves

$$|c_j| \leq \sqrt{1 - e^{-2}} \quad (1 \leq j \leq m) \quad \Rightarrow \quad \sum_{j=1}^m \left(\prod_{k=1}^m |1 - \bar{c}_j c_k| \right)^{1/m} \leq m.$$

To see the mechanism, set $h_j = m^{-1} \sum_k \log |1 - \bar{c}_j c_k|$ and $D_j = -\log(1 - |c_j|^2)$. The logarithmic kernel gives $\sum_j h_j \leq 0$ and $h_j^2 \leq (-m^{-1} \sum_i h_i) D_j$. The radius bound gives

¹The Lean modules `PaperAnalyticClosureAuditR11`, `PaperAnalyticR10`, `PaperAnalyticR10Audit`, `PaperAnalyticR11`, `PaperAnalyticSmokeR11`, `PaperWeightedCoverageR11`, `ThreePointDefectR11` cited in this section are in the author's working tree and are not yet in the public repository; `WeightedBoundaryDeficitR10` and `WeightedCauchyTransportR10` are.

$D_j \leq 2$. Writing $r^2 = -m^{-1} \sum_i h_i$, the resulting bounds $\sum_j h_j = -mr^2$ and $|h_j| \leq \sqrt{2}r$ feed the exponential energy inequality, yielding $\sum_j e^{h_j} \leq m$. Thus the series estimates are proved for the actual configuration, rather than assumed as suppliers. This checks a central equal-weight case; the short paper's ordinary all-disc weighted inequality has greater mathematical scope.

The earlier adaptive certificate remains useful for other configurations.

For $m \geq 1$, let $|c_i|^2 \leq a < 1$, with $a \geq 0$, and put $L = -\log(1-a)$. The [uniform-radius theorem](#) gives the actual geometric-mean inequality

$$e^L \leq 1 + 2L \quad \Rightarrow \quad \sum_{i=1}^m \left(\prod_{j=1}^m |1 - \bar{c}_i c_j| \right)^{1/m} \leq m.$$

Its mechanism is an adaptive logarithmic certificate. Write $H_i = m^{-1} \sum_j \log |1 - \bar{c}_i c_j|$ and $D_i = -\log(1 - |c_i|^2)$. For nonnegative caps $M_i \geq H_i$, the [checked adaptive theorem](#) requires only $\sum_i \Phi(M_i) D_i \leq m$, where $\Phi(t) = \int_0^1 (1-s)e^{st} ds$. The uniform radius gives $H_i, D_i \leq L$, so $\Phi(L)L \leq 1$ suffices; the displayed exponential condition is exactly this inequality when $L > 0$, and $L = 0$ is immediate.

Four points without a matching or numerical split. The preceding weighted proof with $m = 4$ and $w_j = 1/4$ gives

$$\sum_{j=1}^4 \left(\prod_{k=1}^4 |1 - \bar{c}_k c_j| \right)^{1/4} \leq 4, \quad |c_j| \leq 1,$$

with equality if and only if all four centres vanish. The checked four-point theorem states both conclusions, and the strict form retains strictness as soon as a centre is nonzero, including repeated centres and unit-circle centres.

The former 21/25 central/outer split, its perfect-matching Hölder step, and its polynomial, Taylor, logarithmic and radical-calculus certification claims are removed from this record. A scalar stationary-point inequality alone does not establish that whole route. The central-region and adaptive proofs immediately above remain independent, restricted results, not certificates for the removed four-point estimates. The replacement weighted proof and its explicit three- and four-point endpoints pass their source-bound compilation and named axiom checks.

The budget guarantees a critical value at most R^n , but does not force the smaller quintic threshold $1/M_5$ below. On the unit radial family all critical values have modulus one, so a uniformly strict improvement is unavailable. Nor does the value budget bound inverse-ray lengths: a path selection or metric comparison is still needed.

The degree-five target is now explicit

The degree-five analysis sharpens the two-nearest-spoke threshold. If a critical point c satisfies $|f(c)| \leq 1/M_n$, then both straight segments to the two nearest roots remain in $\{|f| \leq 1\}$ and their total length is at most 2. The envelope is obtained by maximising the

exact product along the second spoke; at degree five,

$$M_5 = (1 - t_*)(1 + t_*)^3 \sqrt{16t_*^2 - 4t_* + 1}, \quad t_* = \frac{5}{16} + \frac{3\sqrt{105}}{80}, \quad \frac{1}{M_5} = 0.2760461 \dots$$

This is an 11-fold improvement over the earlier deep-low threshold, but it does not assert that every quintic has a critical value below this level.

The exact remaining degree-five statement is therefore not “improve the constant” but the following selection problem:

For every monic quintic with roots in the closed unit disc, there is a critical point c with $|f(c)| \leq 1$ and two roots a, b such that $|f| \leq 1$ on $[c, a] \cup [c, b]$ and $|c - a| + |c - b| \leq 2$.

The public file calls this (SPOKE-5). Together with the existing two-segment reduction it would settle degree five, but (SPOKE-5) is not proved. Coverage experiments put the residual in rapid, nearly simultaneous merging near the regular pentagon: the generic sampled families were covered, whereas the surviving band has every named merge threshold active and D near 1. Those percentages locate the work; they are not a theorem.

At the same degree, the terminal connected component admits a sharper area identity. If $K_t = \{|f| \leq t\}$ is connected and the centred exterior map is

$$\psi(\zeta) = t^{1/n}\zeta + a_0 + \sum_{k \geq 1} a_k \zeta^{-k},$$

then Grönwall’s identity gives

$$\text{Area}(K_t) = \pi \left(t^{2/n} - \sum_{k \geq 1} k |a_k|^2 \right), \quad a_1 = -\frac{c_{n-2}}{n t^{1/n}} \quad \text{after centring.}$$

It replaces a Pólya upper bound at that terminal node, but extending it to proper components and connecting it to the parent theorem remains open. The degree-five statements and their evidence classes are recorded in [degree5 assembly and sharpened cuts](#).

The no-go boundaries force hub selection

The same frontier is useful because it closes off three misleading readings. First, the strict-argmin first-merge hub need not have two-arm length below 2. A degree-five tie-locus witness, moved into the open unit disc, has

$$L(c_*) = 2.0573432753 \dots > 2,$$

with a 50-digit two-instrument reconstruction and a margin about 97 times that of the earlier degree-four witness. This is computational certificate evidence, not an exact rational theorem; the parent remains viable because a different hub at the same configuration supplies the relevant pair.

Second, the aggregate estimate $\sum_c L(c) \leq 2(n-1)R$ is refuted at degree four on an open violating region and at one degree-five witness. Its algebraic factor

$$\sum_{k=1}^{n-1} |f(c_k)|^{1/n} \leq (n-1)R$$

is proved for every $n \geq 2$ in Theorem 8.1. The metric factor is not automatic in any of these degrees: the all-degree value estimate supplies no path-length aggregate. These two boundaries, including the distinction between the surviving algebraic budget and the failed length aggregate, are in [minimal hub arm budget refutation](#) and [separatrix aggregate reduction](#).

Third, the origin is not a uniform near-Fekete hub. An exact rational quintic with roots scaled by 999999/1000000 has four of five origin spokes escaping $\{|f| \leq 1\}$ at explicit rational sample points, even though $1 - D = 3.19991 \cdot 10^{-4}$. This witness rules out the origin-spoke selector on every defect range that contains its value. Ruling the selector out on every positive neighbourhood of the Fekete locus would need a sequence of witnesses with defect tending to zero, or a limiting argument proving that assertion; neither is supplied here. The mechanism is angular: the first several Fourier modes can point against different spokes. The certificate is exact for the displayed witness; the claim that the phenomenon persists throughout a limiting family is computational. The witness was engineered against the origin selector, and a different admissible hub at the same configuration supplies a contained pair. What fails here is the origin selector; the admissible hubs are not exhausted. The full witness and replay route are in [near fekete radial angular split](#).

The surviving carrier and the actual open endpoint

These refutations leave a selector as the surviving carrier. On the ray-separated dense class, the public frontier records the canonical open row

$$\min_{c \text{ admissible}} L(c) \leq 2,$$

where admissibility includes the two inverse-ray arms being contained in the relevant lemniscate component. Its measured values remain below 2 at the refuting configurations, and the existing attachment and lower semicontinuity reductions would turn this row into the parent theorem. No proof of the row is currently available. In particular, the nonzero hub that rescues the origin-spoke witnesses is evidence for the selector, not a universal construction.

The honest current endpoint is therefore a containment selector with three separate obligations: choose the hub, keep both arms inside the component, and control the metric fan-in. The Lean module cited elsewhere in this note checks the Newton value equation, ray-collision avoidance, and root-retention inputs; the new frontier files above are ordinary proof, exact-certificate, and computational research records, not formal authority for (SPOKE-5), the no-go generalisations, or the parent theorem. Keeping those authority boundaries visible is part of making the frontier reusable.

9 The Newton value equation

Away from the critical set, define the complex Newton field

$$N(z) = -\frac{f(z)}{f'(z)}.$$

Let $I \subseteq \mathbb{R}$ be an interval. On the interval under consideration, assume z is differentiable, $f'(z(t)) \neq 0$, and $z'(t) = N(z(t))$; put $w(t) = f(z(t))$. Kozen and Stefánsson record the following identity as a lemma of Shub, Tischler and Williams [7, Lemma 2.1].

Theorem 9.1 (value equation). *Let f be a polynomial and $z : I \rightarrow \mathbb{C}$ a differentiable curve on an interval I , with $f'(z(t)) \neq 0$ and $z'(t) = -f(z(t))/f'(z(t))$ throughout I . For $w = f \circ z$, one has $w'(t) = -w(t)$ on I .*

The computation is one line: $w' = f'(z) z' = f'(z) \cdot (-f(z)/f'(z)) = -f(z) = -w$. The kernel checks the local complex-parameter chain rule as [newton flow value has deriv at](#), together with the differential form of the first integral, [newton flow scaled value has deriv at zero](#):

$$\frac{d}{dt}(e^t f(z(t))) = 0, \quad \text{equivalently} \quad f(z(t)) = e^{-(t-t_0)} f(z(t_0)), \quad t_0 \leq t, \quad t_0, t \in I.$$

Integration on the real interval gives the last identity as an ordinary consequence of the differential equation. For an existing trajectory with nonzero initial value, its value moves inward on one positive ray; a zero value has no argument and remains zero. Thus $|f| < 1$ is preserved for later times in that trajectory's existing interval. No global existence or description of a whole ray preimage follows from this scalar equation. The real-time candidate endpoint is [newton flow real value decay](#) in [newton flow trajectory](#).

Corollary 9.2 (ray separation). *Let $a < b$ and let the value trajectory $t \mapsto f(z(t))$ be continuous on $[a, b]$. Assume the Newton equation and $f'(z(t)) \neq 0$ on (a, b) only. Then*

$$f(z(b)) = e^{a-b} f(z(a)).$$

If these endpoint values are nonzero, they lie on one positive ray. Therefore critical points with values on distinct positive rays cannot be endpoints of such a finite connection. The trajectory in the z plane need not be radial.

The interior identity passes to the endpoints by continuity, not by evaluating $-f/f'$ at a critical endpoint. The candidate declarations [value decay at continuous endpoints](#) and [no finite connection of distinct value rays](#) in [newton endpoints](#) make these premises explicit. Their compilation and axiom checks are not reported here. They construct no trajectories and supply no global monodromy theorem. Kozen and Stefánsson draw the same ray consequence for the Newtonian graph: under f , every edge maps onto a segment of a ray through the origin whose endpoints are 0 or critical values [7, §2].

10 Arguments, not moduli

It is tempting to arrange a generic perturbation so that the critical values are pairwise distinct, or that their moduli are pairwise distinct, and to conclude that saddle connections are excluded. Neither is enough.

Two distinct critical values can lie on one ray, and two critical values with distinct moduli certainly can: the ray records the argument, and the modulus is exactly the coordinate the flow contracts. By Corollary 9.2 the invariant that excludes connections is the argument. What a perturbation must therefore achieve is pairwise distinct critical-value *arguments*, which is a condition on $n - 1$ points modulo the circle. Their positions in the plane are unconstrained by it.

The cost of that condition is also checked, and it is small.

Theorem 10.1 (ray-collision locus). *Let $a \neq b$ be complex. Every common translation β for which $a + \beta$ and $b + \beta$ lie on the same positive ray has the form*

$$\beta = \frac{ra - b}{1 - r}, \quad r \in \mathbb{R}_{>0}, r \neq 1.$$

So each pair of critical values contributes a one-real-parameter forbidden locus in the translation plane, given in closed form ([translated same positive ray parameterization](#)). A finite union of such loci has empty interior, which is the shape one wants for an avoidance argument. Turning that into a perturbation of f is not immediate: the translation model must be replaced by an actual perturbation of the roots that keeps them inside $\mathbb{D}$ and preserves the length slack. That is the first open producer of §13.

11 A proof gap in the unrestricted argument

The March manuscript's Proposition 12 claims the following load-bearing statement. For $u = -\log|f|$, a connected component V of $\{u > c\}$ carrying $m \geq 2$ simple zeros, and the stated regularity and Morse hypotheses, it constructs, for every $\varepsilon > 0$, an embedded spanning tree $G_\varepsilon \subset V$ with

$$\text{len}(G_\varepsilon) \leq \frac{1}{2\pi} \int_{2\alpha}^\infty P_V(t) dt + \varepsilon. \quad (5.1)$$

The final theorem uses this estimate, so the issue below cannot be bypassed by calling the proposition auxiliary.

The estimate itself is false. Take the Cassini polynomial $f_a(z) = z^2 - a^2$ at $a = 9/10$. Its component contains the two roots at distance $9/5$ and satisfies the relevant Morse and critical-value hypotheses. Direct level-length calculation gives the upper majorant

$$4(\sqrt{a^2 + a} - a) < \frac{41}{25}$$

for the right side of (5.1) before ε , whereas every connected set containing both roots has length at least $9/5$. Since $9/5 - 41/25 = 4/25$, choosing a smaller positive ε contradicts the assertion. Thus neither a different local saddle neighbourhood nor a perfect topological decomposition can recover the printed coefficient $1/(2\pi)$. The

exact inequality and its abstract tree-budget contradiction are checked in `ErdosProblems.Erdos1041.CassiniTreeBudget`; the level-length majorant is an explicit analytic input, not a kernel-checked integral evaluation.

At an interior index-one critical point p , the proof invokes a Morse chart

$$u = \mu + x^2 - y^2$$

and replaces the saddle by a three-ended neighbourhood having one connected lower cross-section and two connected upper cross-sections. That local model is false as written. Because $p \in V$ and V is open, a sufficiently small closed disc around p lies entirely in V . In that disc the full Morse chart has four sectors: two components of $u > \mu$ and two components of $u < \mu$. The level set $\{u = \mu\}$ is a lemniscate graph with a vertex of degree four at p [20, Definition 1.2 and Proposition 2.4]. A global component argument cannot delete one local sector from a disc already contained in V .

This independently diagnoses a proof step, but the Cassini witness above also refutes the proposition's printed metric statement. A different route might cut an adjoining regular annulus along a separatrix or regular flow arc before forming the block, retain a four-pronged saddle neighbourhood and change the assembly, or replace the local construction by the ray-cut decomposition proposed below. But no repair can retain (5.1); it must pay a positive attachment cost, select only one short pair instead of spanning every root, or use a different global metric inequality. The shorter descriptions of the same three-ended block do not repair the four-sector topology.

There is nevertheless a complete ordinary topology theorem on the generic stratum where the critical values separate both by angle and by level. Its ingredients are classical: the Newtonian graph of Shub, Tischler and Williams [7, §2], the component-wise Riemann–Hurwitz count in the proof of [8, Proposition 2.1], and the division of the Riemann surface of the inverse function into sheets by slits along rays from its branch points, which Dubinin uses to estimate capacities [9, §2]. The theorem below specifies the sheets, critical transpositions and descent arcs for the stated component.

Theorem 11.1 (ray-separated slit-sheet decomposition). *Let f be monic, and let U be a component of $\{|f| < 1\}$ containing $k \geq 2$ roots, counted with multiplicity. Suppose every critical point of f in U is simple, its critical value is nonzero, and these critical values have pairwise distinct arguments and pairwise distinct moduli. Then:*

1. $-\log |f| : U \setminus f^{-1}(0) \rightarrow (0, \infty)$ is a proper excellent Morse function, with exactly $k - 1$ nondegenerate saddles;
2. cutting the value disc along the complete critical rays decomposes the preimage into conformal strips;
3. cutting each ray only from its critical value to the unit circle gives k conformal sheets, one per root, whose critical transpositions form a tree;
4. for each critical point c , the two inverse lifts of $[0, f(c)]$ join two roots through c inside $\{|f| \leq |f(c)|\}$, and the union of these arcs is an embedded geometric realisation of that tree.

Small neighbourhoods of the saddles can be chosen with diameter $O(\sqrt{\delta})$ at value radius δ .

Proof. A polynomial sublevel component is simply connected, and $f : U \rightarrow \mathbb{D}$ is proper of degree k . Riemann–Hurwitz gives exactly $k - 1$ simple critical points. The holomorphic Morse lemma makes them nondegenerate saddles of $-\log|f|$; the distinct moduli make their levels distinct. Properness here is for the displayed codomain $(0, \infty)$: a compact positive range stays away from both the level-one boundary and the deleted roots. Removing all critical rays leaves simply connected sectors without branch values, so every component of their preimage maps biholomorphically to its sector; logarithmic polar coordinates turn these components into strips.

For the sheet tree, remove only the outward slits $\{tf(c) : 1 \leq t < 1/|f(c)|\}$. Distinct critical-value arguments make the remaining value domain star-shaped about 0, hence simply connected. Its preimage splits into k sheets labelled by the roots. Local monodromy at a simple critical value is a transposition. The monodromy action before the slits are made is transitive because the punctured domain upstairs is connected; therefore the graph of the $k - 1$ transpositions is connected. A connected graph with k vertices and $k - 1$ edges is a tree.

The inward segment from $f(c)$ to 0 contains no other critical value: distinct arguments exclude a different ray, and distinct moduli separate critical levels on the same ray. Its two inverse lifts converge to c by the holomorphic Morse coordinate and start at two roots. Lifts belonging to different critical points cannot cross away from roots, since an intersection would give one nonzero value on two critical rays. Their union is therefore an embedded tree. The same local Morse coordinate identifies a value disc of radius δ with a spatial neighbourhood of diameter $O(\sqrt{\delta})$. $\square$

The complete ordinary argument is recorded in [the attachment-aware Reeb source](#). Its Lean companion checks the ray-distance, averaging and finite-tree kernels; it explicitly does not formalise Riemann–Hurwitz, monodromy, the strip diffeomorphisms or the geometric assembly. The theorem supplies topology, not a useful length sum: the canonical inverse-ray tree can already have the Cassini deficit discussed above.

Corollary 9.2 shows that the distinct-argument hypothesis also excludes finite non-singular saddle-to-saddle Newton connections. The theorem does not cover coincident arguments, simultaneous critical levels, multiple saddles, boundary tangencies in a general compact band, or non-Hausdorff orbit spaces, and it gives no target-scale metric gluing estimate. Those broader questions remain below.

12 Finite evidence

A search was run over random monic polynomials with roots in the unit disc. For each sample the region $\{|f| < 1\}$ was rasterised and shortest grid paths were computed between every pair of roots. The best upper bounds obtained were

degree 5,	500 trials :	1.1052648928
degree 6,	1500 trials :	0.8450414343
degree 8,	1500 trials :	0.6203916714
degree 10,	1500 trials :	0.4303640486.

No counterexample candidate was found, and the measured values sit well below the threshold 2. These are numerical candidate connectors. A polygon whose vertices satisfy $|f| < 1$ need not lie in the open lemniscate, since an edge can cross the boundary between two safe vertices, and root snapping adds further edges of the same kind. Turning a candidate into a proof needs a continuous certificate on every edge, for instance strict positivity of the real polynomial $1 - |f(z(t))|^2$ on $[0, 1]$ for each straight edge $z(t) = u + t(v - u)$, established by exact coefficients or outward-rounded interval arithmetic with subdivision, together with exact root enclosures. No such certificate is attached to the numbers above. They are grid distances for the sampled configurations, and the apparent decrease with degree is a property of the sample, from which we draw no conjecture. Even a finite collection of proved instances would leave the universal statement open. Future searches should target four named failure modes: near-degenerate saddles, thin necks, boundary-critical configurations, and almost-connected separatrices. They should report those diagnostics. Raster paths remain candidate finders, never continuous certificates.

13 Complements and further questions

The dependency chain has five separate gates. A proof, a minimally corrected hypothesis set, or an explicit polynomial or planar counterexample is a decisive answer to any one of them.

1. Repair or refute the saddle block

Problem 13.1 (corrected local saddle assembly). The printed $1/(2\pi)$ spanning-tree estimate of Proposition 12 is false: the Cassini polynomial $z^2 - a^2$ at $a = 9/10$ makes the proposed tree budget strictly shorter than the distance between its roots. Any later argument must pay a positive attachment cost, select only one short pair instead of spanning every root, or use a different global metric inequality. A four-pronged or cut-annulus local model may still be useful for another estimate, but it cannot recover (5.1).

The full-disc model $x^2 - y^2$, an annulus in which lower branches rejoin, two saddles joined by a separatrix and simultaneous saddle levels are mandatory tests. Repeating the one-lower/two-upper assertion does not answer the problem.

2. The compact ray-cut decomposition

Let p have simple roots, simple nonzero critical points and let $u = -\log |p|$. For regular values $c < T$, take

$$M_{c,T} = \overline{V \cap \{c < u < T\}},$$

and assume explicitly that this is a compact genus-zero surface, its lower boundary is one smooth Jordan curve, its upper boundary consists of m smooth root curves, all interior critical points are nondegenerate index-one saddles, the normalised gradient field

$$X = \frac{\nabla u}{|\nabla u|^2} = -\frac{p}{p'}$$

is transverse to the level boundaries, and no maximal X -trajectory has two saddle end-points.

Problem 13.2 (finite strip decomposition after ray cuts). For every $\eta > 0$, construct disjoint Morse neighbourhoods N_j with $\sum_j \text{diam } N_j < \eta$ and a finite set of complete separatrix or regular-flow cuts so that every component of the complement is flow-diffeomorphic to a rectangle

$$[a_S, b_S] \times [0, 1], \quad u(\Phi_S(t, s)) = t,$$

with connected level sections and no uncut annular component. Give an explicit finite bound $E(s, m)$ for the number of strips, or exhibit the minimal missing hypothesis or a counterexample.

No particular formula such as $2s + 1$ is presumed. Boundary tangencies, simultaneous levels, branch reunion through an annulus and non-Hausdorff orbit spaces must all be handled explicitly.

3. Metric fan-in after the printed spanning-tree coefficient is excluded

For a strip S , write

$$\Gamma_t^S = S \cap \{u = t\}, \quad P_S(t) = \mathcal{H}^1(\Gamma_t^S),$$

and let k_S be its number of root ends. Write

$$\Phi_S(t) = \int_{\Gamma_t^S} |\nabla u| ds$$

for its transverse flux. Normalising the transverse measure by Φ_S gives the average trajectory estimate

$$\int_{\Gamma_{t_0}^S} \text{len}(\gamma_x) d\mu_{t_0}(x) = \frac{1}{\Phi_S(t_0)} \int_{a_S}^{b_S} P_S(t) dt.$$

The flux of a strip is not an integer multiple of 2π . For $f(z) = z$ and $u = -\log |z|$, an annular sector of angular width θ has $|\nabla u| = 1/r$ and $ds = r d\phi$ on a circular level arc, so $\Phi_S = \theta$; for z^n the same computation gives $n\theta$. A closed level curve enclosing roots does carry an integer winding flux, and cutting a regular annulus destroys that. Those identities explain why a per-strip average does not automatically yield a global coefficient. They do not reopen (5.1). Cassini already excludes the printed coefficient $1/(2\pi)$ for any embedded tree that contains all m roots: Problem 13.1 records that no later gluing argument can retain that estimate.

Problem 13.3 (strip gluing after the printed coefficient). Assuming Problem 13.2, do one of the following, for every $\eta > 0$.

1. Produce an embedded tree G containing all m roots whose length obeys

$$\text{len}(G) \leq C \int_{2\alpha}^{\infty} P_V(t) dt + \eta \tag{6.1}$$

for an explicit constant $C > 1/(2\pi)$ that accounts for attachment cost and for a valid allocation of the strip fluxes Φ_S .

2. Connect only one pair of distinct roots, rather than spanning every root.
3. Give a different global metric inequality that is not (5.1).

The printed coefficient $1/(2\pi)$ is not an open target. Better additive control of saddle, annular-cut and root-cap cost cannot recover (5.1).

The printed collar slack

$$q = \frac{1}{2\pi} \int_{\alpha}^{2\alpha} P_V(t) dt > 0$$

used the same excluded coefficient. It is not remaining budget on an otherwise complete spanning-tree estimate. Independently choosing a shortest trajectory in each strip is not enough unless the attachment mismatch is controlled, and controlling that mismatch does not restore the printed $1/(2\pi)$ tree bound.

4. Coefficient perturbation and stability

The constant-translation stage is no longer open. Once a finite critical-value family is injective, Lean proves an arbitrarily small translation making every value nonzero and pairwise positive-ray separated ([exists small translation separating arguments](#)). It also proves the explicit root-retention estimate; a shift below ε keeps all roots in the unit disc when

$$((n+1)\varepsilon)^{1/n} + \rho < 1$$

([constant perturbation roots in unit disk](#)). A constant translation cannot separate initially equal critical values.

Problem 13.4 (two-stage generic perturbation with slack). For fixed root discs, compact collar K , regular levels $\alpha/2, \alpha, 2\alpha, 5\alpha/2$ and collar slack $q > 0$, prove that an arbitrarily small

$$g_{\lambda,\beta}(z) = f(z) + \lambda z + \beta$$

can be chosen so that:

1. $f + \lambda z$ has simple relevant critical points and injective complex critical values;
2. the checked constant translation β makes those values nonzero and pairwise ray-separated;
3. no critical point enters the protected collar or truncation boundary;
4. the relevant component remains in K , contains exactly the corresponding perturbed roots and has slack $q_g > q/2$; and
5. root displacement and straight-line transfer back to the original roots consume less than a prescribed fraction of q/m .

If the one-coefficient perturbation λz cannot ensure all five properties, give the smallest additional lower-coefficient direction that can, or an explicit obstruction.

Finite planar avoidance and the subsequent constant translation must not be relisted as missing; coefficient genericity, component stability and slack stability are the open content.

5. The global Newton-flow claim ceiling

Classical Newton-flow theory supplies the global part of this programme. Away from the roots and critical points, every orbit satisfies

$$p(z(t)) = e^{-(t-t_0)} p(z(t_0)), \quad u(z(t)) = u(z(t_0)) + t - t_0, \quad u = -\log |p|,$$

by the lemma of Shub, Tischler and Williams [7, Lemma 2.1]; every maximal orbit has one of four types, with finite ends at critical points, an infinite forward end at a root and an infinite backward end at the point at infinity [7, Lemma 2.2]; and the orbits joining roots and critical points form a connected Newtonian graph with finitely many edges [7, Definition 2.1 and §2].

Problem 13.5 (relative global Newton-flow theorem). On a compact regular band

$$M_{a,b} = \{z : a \leq -\log |p(z)| \leq b\},$$

adapt this theory to the band: describe how maximal orbits cross the two boundary curves, and show that the orbit space of the chosen strip decomposition is Hausdorff. Under pairwise ray separation of the critical values, decide whether the flow is Morse–Smale relative to the boundary or whether the strongest valid conclusion is only absence of saddle-to-saddle connections.

The checked algebra proves the pointwise value equation and the endpoint-ray consumer; the global solution theory and the finite graph are the classical results cited above. The open content is their relative form on the band, together with the quantitative slack and length control needed for gluing. A negative answer should exhibit the simplest ray-separated polynomial carrying the remaining pathology.

Erdős #1041 remains open. Five separate statements are proved above, with different degree ranges, root hypotheses, containment levels and constants. Theorem 3.1 holds in every degree $n \geq 2$ for squarefree f under the single hypothesis $\mu \leq 13/25$, with no condition on where the roots lie, and gives a connector inside $\{|f| < 1\}$ of length below 2; the regime $13/25 < \mu < 1$ is untouched. Theorem 4.1 drops every threshold and every root hypothesis, and pays for that with the constant 71/10 and the weaker containment level $\{|f| \leq 2\mu\}$; the containment $\{|f| < 1\}$ with length at most 5.7 needs only $\mu \leq 1/2$, and both bounds fall short of the target constant 2. Theorem 5.1 settles degree three completely for roots in the open unit disc. Theorem 6.1 holds in every degree and is conditional on the critical-value separation S , with the uniform threshold $S = 4/3$ reaching every degree $n \geq 3$ and every real centre of the normalised value segment by Corollary 6.2. Theorem 2.1 settles the whole trinomial family in every degree by an ordinary radial argument. Their formal coverage differs: the translated cubic fibres have a complete rectifiable-path theorem; the trinomial source checks both radial bounds and the radius sum. The remaining companions check the named numerical, algebraic and power-series steps inside them, the degree-three companion lies outside the pinned formal-source library and carries no kernel receipt in this release, and the disk uniformisation, Bergman estimate, capacity gap, area bounds and certificate chain remain ordinary mathematics. The source now publicly verifies the Newton kernel, finite ray avoidance and quantitative constant-translation root control. The ray-separated

generic stratum has the ordinary slit-sheet decomposition of Theorem 11.1; coefficient perturbation, the exceptional compact-band decompositions and metric gluing remain unresolved.

Statements and declarations

Lean does not check the exposition, citation choices, or interpretation. This manuscript cites Lean only for the formal statements and proofs that the pinned kernel accepts. The checked core is the local Newton value equation, the zero-derivative form of its exponentially scaled value, the algebraic consumer from an assumed exponential endpoint relation, the finite planar-avoidance theorem, quantitative constant-translation root retention, and the ray-collision parameterisation. The real-time first integral and finite-trajectory ray exclusion use the ordinary interval-integration and endpoint-continuity argument given in the paper. The decomposition and length statements of §13 are not proved. The diagnosis of Proposition 12 in §11 records both the invalid local saddle block and the Cassini obstruction to the printed global tree budget. The search results of §12 are computations.

A Guide to the formal sources

The public `ErdosProblems.Erdos1041.NewtonFlowRaySeparation` module contains the checked source for this note. The search of §12 is `scripts/search_counterexample.py` in the source package. The declaration table below is pinned to the shared formal-source commit used throughout this problem-note series.

- `newton flow vector`
- `derivative mul newton flow vector`
- `newton flow value has deriv at`
- `newton flow scaled value has deriv at zero`
- `same positive ray`
- `same positive ray refl`
- `same positive ray symm`
- `same positive ray trans`
- `translated same positive ray parameterization`
- `real affine line`
- `real affine line eq affine span`
- `is closed real affine line`
- `dense compl real affine line`
- `exists small avoiding finite real affine lines`

- same positive ray imp mem real affine line
- exists small translation separating arguments
- norm lt one of near root
- constant perturbation root near original
- constant perturbation roots in unit disk
- same positive ray of real exp decay
- no newton connection of not same positive ray

Checked declarations behind the translated quotient-fibre families. For the translated cubic and quartic quotient families $f(z) = P((z - h)^q)$ the metric theorems are ordinary; the kernel checks the cyclic-fibre mean square and the quotient-disk bridge. The fibre norm identity is [fibre mean square](#), the strict unit bound is [fibre unit bound](#), the quotient roots land in the open disk by [quotient disk bridge](#), and the composition vanishes on the fibre by [fibre zero identity](#).

B Record: the circle-slice arity floor in numbers

This part is long-record only. Nothing in the short note depends on it, and none of it is a certified upper bound. It records what the arity floor behind Theorem 3.1 is worth, so that a reader deciding whether to attack the complementary regime $13/25 < \mu < 1$ can see the size of the remaining slack.

At $a = 1$ the constants of the failure inequalities are $\delta = 0.4586751$, $D = 5.3770727$, $d_{\text{low}} = 2.1700771$, and $\cosh(D/2) = e^2$. Four columns are worth comparing at that area: the earlier corpus profile $\Lambda(k, 1) = \max(\delta/2 + (k-1)\lambda(g), k\tau/2)$ with $\tau = 0.2723415$; the hyperbolic packing floor Λ_{pack} that the previous threshold $2/5$ was built on; the linear-programming value M_{circ} of the relaxation of Lemma 3.3; and the best configuration a floating optimiser could realise.

k	corpus Λ	packing	M_{circ}	realisable
2	0.3103	0.5899	0.3102	0.3103
3	0.4085	0.6878	0.3893	0.3784
4	0.5447	0.7475	0.4450	0.4306
5	0.6809	0.7906	0.4846	0.4694
6	0.8170	0.8244	0.5155	0.4986
7	0.9532	0.8521	0.5408	0.5208
8	1.0894	0.8757	0.5622	0.5413
10	1.3617	0.9142	0.5972	0.5759
14	1.9064	0.9710	0.6487	0.6228
20	2.7234	1.0298	0.7022	0.6717
30	4.0851	1.0955	0.7624	0.7283
100	13.617	1.2871	0.9428	—

Both right-hand columns are floating searches, and the M_{circ} column is additionally a grid value on a 2600-point d -grid and a 1400-point r -grid, so it is a lower bound for the relaxation. The $k = 2$ row shows the size of that artifact, 0.31018 against the exactly known 0.310338. Their proximity suggests a small gap and gives no rigorous bracket.

The certified floors themselves, as integers after rounding up, read as follows at $a = 1$.

x	0.40	0.45	0.50	0.55	0.60	0.63	0.65	0.70
corpus profile	3	4	4	5	5	5	5	6
packing floor	2	2	2	2	2	3	3	4
circle-slice floor	3	4	6	8	10	12	14	18

In one explicit-Euler floating replica with step 10^{-3} and a geometric grid of 40 initial areas from 10^{-6} to 1, the supremum of the hitting time falls from 0.89703 with the packing floor to 0.65503 with the certified circle-slice pairs, and to 0.63003 if the relaxation's own linear-programming value replaces the certified pairs. Shrinking the relaxation by the worst measured slack factor 1.045 moves it only to 0.60703. These computations suggest a mechanism ceiling near $e^{-0.607} = 0.545$; they prove no such ceiling, and they exclude neither a stronger arity bound nor a larger certified regime.

Two smaller leaks in the present certificate are engineering. The dual is restricted to p radii, ten in quick mode and fourteen in full, which at $a = 1$, $k = 8$ costs about one per cent against the unrestricted family; the a -grid is coarse and rounded up, at a comparable cost.

One sub-statement is genuinely open and is smaller than the parent problem. Determine $M_{\text{circ}}(k, a)$ exactly, that is, maximise $\sum_j \lambda(d_j)$ subject to $d_j \geq d_{\text{low}}$ and $\sum_j w(d_j, r) \leq$

π for every $r > 0$. The optimiser's solution is a spread-out density in d , and it appears to grow like $(\pi / \sinh(D/2)) c \log k$. A closed form would determine the relaxation and clarify the geometry of the failure configuration. How much it would improve the final threshold is unproved.

C Complete result-family map

This section places every registered family for this problem in the shared 70-family reader order. The five display bands control exposition only; the separate promotion state currently covers 6 families and is reported but does not hide or strengthen any family. Mathematical statements and evidence modes come from the public claim registry. Across all eight problems the public result-atom catalog contains 681 exact packet coordinates; this problem contributes 182. The recovered source catalog contains 682 rows; 1 row(s) belong to families absent from the current claim registry and remain disclosed as detached source rows. Catalog rows expose bounded statement excerpts plus full-source digests, not a claim that every complete packet statement is reproduced here.

C.1 Root retention

Reader position. 7 of 70; display band: front door. Formal editorial disposition: split. These are separate classifications.

Reader entry. A quantified small constant perturbation keeps every polynomial root inside the open unit disc.

A quantified small constant perturbation keeps every polynomial root inside the open unit disc.

Authority and reach. Lean kernel; Comparator-selected; locally proved result; novelty unassessed.

Exact boundary. Root retention is one input to a still-incomplete route.

Result-atom population. 51 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. exact selected interface; 1 executable interface(s), 1 repository-registered selected result interface(s).

Publication placement. This family is also admitted to the short note.

C.2 Newton value decay

Reader position. 13 of 70; display band: major result. Formal editorial disposition: split. These are separate classifications.

Reader entry. Lean checks the local Newton value equation and the zero-derivative form of its exponentially scaled value; constancy along a real-time interval is an ordinary calculus consequence in the paper.

Lean checks the local Newton value equation and the zero-derivative form of its exponentially scaled value; constancy along a real-time interval is an ordinary calculus consequence in the paper.

Authority and reach. Lean kernel plus authored synthesis; locally proved result and formalised calculus; novelty unassessed.

Exact boundary. The cited declarations are pointwise derivative statements. The real-time first integral requires the paper's interval integration step, and value decay alone does not construct a global trajectory or give a short connecting curve.

Result-atom population. 95 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. not selected for comparator; 0 executable interface(s), 0 repository-registered selected result interface(s).

Publication placement. This family is also admitted to the short note.

- `ErdosProblems.Erdos1041.newtonFlow_value_hasDerivAt`
- `ErdosProblems.Erdos1041.newtonFlow_scaledValue_hasDerivAt_zero`

C.3 Ray separation

Reader position. 25 of 70; display band: mechanism. Formal editorial disposition: split. These are separate classifications.

Reader entry. Lean checks that distinct positive rays contradict an assumed exponential endpoint relation; the paper derives that relation for a finite real-time Newton trajectory by ordinary integration and endpoint continuity.

Lean checks that distinct positive rays contradict an assumed exponential endpoint relation; the paper derives that relation for a finite real-time Newton trajectory by ordinary integration and endpoint continuity.

Authority and reach. Lean kernel plus authored synthesis; locally proved result; novelty unassessed.

Exact boundary. The finite-trajectory conclusion retains continuity on the closed interval, differentiability and noncritical Newton evolution on its interior, and the ordinary endpoint passage. It supplies neither global trajectory existence nor geometric length control.

Result-atom population. 19 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. represented by selected interface; 0 executable interface(s), 0 repository-registered selected result interface(s).

Publication placement. This family is also admitted to the short note.

- `ErdosProblems.Erdos1041.samePositiveRay_of_real_exp_decay`
- `ErdosProblems.Erdos1041.no_newtonConnection_of_not_samePositiveRay`

C.4 Published proof gap

Reader position. 46 of 70; display band: frontier. Formal editorial disposition: split. These are separate classifications.

Reader entry. The paper identifies an invalid local saddle block in a claimed unrestricted proof.

The paper identifies an invalid local saddle block in a claimed unrestricted proof.

Authority and reach. authored source analysis; paper diagnosis.

Exact boundary. The diagnosis does not prove that no repair exists.

Result-atom population. 4 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. not applicable to comparator; 0 executable interface(s), 0 repository-registered selected result interface(s).

Publication placement. This family is also admitted to the short note.

C.5 Lemniscate finite search

Reader position. 55 of 70; display band: frontier. Formal editorial disposition: split. These are separate classifications.

Reader entry. A bounded search records finite geometric evidence.

A bounded search records finite geometric evidence.

Authority and reach. external computation; finite computation.

Exact boundary. Finite evidence does not settle the universal problem.

Result-atom population. 6 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. not applicable to comparator; 0 executable interface(s), 0 repository-registered selected result interface(s).

Publication placement. This family remains in the complete long record and is not a short-note headline.

C.6 Translation avoidance

Reader position. 61 of 70; display band: technical support. Formal editorial disposition: split. These are separate classifications.

Reader entry. Collision translations are parameterised by real affine lines and an arbitrarily small common translation avoids finitely many of them.

Collision translations are parameterised by real affine lines and an arbitrarily small common translation avoids finitely many of them.

Authority and reach. Lean kernel; Comparator-selected; locally proved result; novelty unassessed.

Exact boundary. The finite-family avoidance theorem does not perform the global topology or metric gluing.

Result-atom population. 7 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. exact selected interface; 1 executable interface(s), 1 repository-registered selected result interface(s).

Publication placement. This family remains in the complete long record and is not a short-note headline.

References

- [1] G. Pólya, *Beitrag zur Verallgemeinerung des Verzerrungssatzes auf mehrfach zusammenhängende Gebiete*, Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-Mathematische Klasse (1928), printed pp. 228–232 and 280–282. <https://archive.org/details/sitzungsbericht1928preu>.
- [2] E. Crane, *The areas of polynomial images and pre-images*, Bull. London Math. Soc. **36** (2004), no. 6, 786–792, doi:[10.1112/S0024609304003509](https://doi.org/10.1112/S0024609304003509); preprint arXiv:[math/0302189v1](https://arxiv.org/abs/math/0302189v1), whose statement numbers are cited.
- [3] T. F. Bloom, *Erdős Problems*, problem 1041. <https://www.erdosproblems.com/1041>

- [4] P. Erdős, F. Herzog, and G. Piranian, *Metric properties of polynomials*, J. Analyse Math. **6** (1958), 125–148, doi:[10.1007/BF02790232](https://doi.org/10.1007/BF02790232).
- [5] S. Ghosh and K. Ramachandran, *Number of Components of Polynomial Lemniscates: A Problem of Erdős, Herzog, and Piranian*, J. Math. Anal. Appl. **540** (2024), no. 1, 128571, doi:[10.1016/j.jmaa.2024.128571](https://doi.org/10.1016/j.jmaa.2024.128571); preprint arXiv:[2312.13673v1](https://arxiv.org/abs/2312.13673), whose statement numbers are cited.
- [6] S. Sutherland, *Bad Polynomials for Newton’s Method*, in B. Bielefeld and M. Lyubich (eds.), *Conformal Dynamics Problem List*, Stony Brook IMS preprint (1992), 42–44. <https://www.math.stonybrook.edu/preprints/ims92-7.pdf>
- [7] D. Kozen and K. Stefánsson, *Computing the Newtonian graph*, J. Symbolic Comput. **24** (1997), no. 2, 125–136, doi:[10.1006/jsco.1997.0118](https://doi.org/10.1006/jsco.1997.0118); authors’ copy <https://www.cs.cornell.edu/kozen/Papers/newton.pdf>.
- [8] P. Ebenfelt, D. Khavinson, and H. S. Shapiro, *Two-dimensional shapes and lemniscates*, in *Complex Analysis and Dynamical Systems IV, Part 1*, Contemp. Math. **553**, Amer. Math. Soc., Providence, RI, 2011, 45–59, doi:[10.1090/conm/553/10931](https://doi.org/10.1090/conm/553/10931); preprint arXiv:[1003.4567v1](https://arxiv.org/abs/1003.4567), whose statement numbers are cited.
- [9] V. N. Dubinin, *Lemniscates and inequalities for the logarithmic capacities of continua*, Mat. Zametki **80** (2006), no. 1, 33–37, doi:[10.4213/mzm2777](https://doi.org/10.4213/mzm2777); English translation, Math. Notes **80** (2006), no. 1, 31–35, doi:[10.1007/s11006-006-0105-8](https://doi.org/10.1007/s11006-006-0105-8).
- [10] shtuka, *A Short Path Joining Two Zeros Inside a Polynomial Lemniscate*, manuscript posted 24 March 2026, 48 pp. <https://shtuka123.github.io/1041/main.pdf>. The file at this URL has since been replaced by a shorter partial version that no longer contains Proposition 12; the durable public record of the March version, its defect, and the author’s 26 March 2026 concession is the discussion thread at <https://www.erdosproblems.com/forum/thread/1041>.
- [11] V. S. Pendyala, *A Degree-Four Lemniscate Path Theorem*, arXiv:[2606.24875v1](https://arxiv.org/abs/2606.24875) (2026), doi:[10.48550/arXiv.2606.24875](https://doi.org/10.48550/arXiv.2606.24875).
- [12] V. N. Dubinin, *Some inequalities for polynomials and rational functions associated with lemniscates*, Zap. Nauchn. Sem. POMI **404** (2012), 83–99; English translation, J. Math. Sci. **193** (2013), no. 1, 45–54, doi:[10.1007/s10958-013-1432-4](https://doi.org/10.1007/s10958-013-1432-4).
- [13] P. Borwein, *The arc length of the lemniscate $\{|p(z)| = 1\}$* , Proc. Amer. Math. Soc. **123** (1995), no. 3, 797–799, doi:[10.1090/S0002-9939-1995-1223265-3](https://doi.org/10.1090/S0002-9939-1995-1223265-3).
- [14] A. Eremenko and W. Hayman, *On the length of lemniscates*, Michigan Math. J. **46** (1999), no. 2, 409–415, doi:[10.1307/mmj/1030132418](https://doi.org/10.1307/mmj/1030132418); preprint arXiv:[0805.2295](https://arxiv.org/abs/0805.2295).
- [15] A. Fryntov and F. Nazarov, *New estimates for the length of the Erdős–Herzog–Piranian lemniscate*, in *Linear and Complex Analysis*, Amer. Math. Soc. Transl. Ser. 2 **226**, Amer. Math. Soc., Providence, RI, 2009, 49–60, doi:[10.1090/trans2/226/05](https://doi.org/10.1090/trans2/226/05); preprint arXiv:[0808.0717v1](https://arxiv.org/abs/0808.0717).

- [16] T. Tao, *The maximal length of the Erdős–Herzog–Piranian lemniscate in high degree*, arXiv:[2512.12455v1](#) (2025).
- [17] A. Eremenko and P. Yuditskii, *Comb functions*, Contemp. Math. **578** (2012), 99–118, doi:[10.1090/conm/578/11472](#); preprint arXiv:[1109.1464v1](#).
- [18] A. Eremenko and L. Lempert, *An extremal problem for polynomials*, Proc. Amer. Math. Soc. **122** (1994), no. 1, 191–193, doi:[10.1090/S0002-9939-1994-1207536-1](#).
- [19] A. Eremenko, *A Markov-type inequality for arbitrary plane continua*, Proc. Amer. Math. Soc. **135** (2007), no. 5, 1505–1510, doi:[10.1090/S0002-9939-06-08640-0](#); preprint arXiv:[math/0606745v1](#).
- [20] C. J. Bishop, A. Eremenko, and K. Lazebnik, *On the shapes of rational lemniscates*, Geom. Funct. Anal. **35** (2025), no. 2, 359–407, doi:[10.1007/s00039-025-00704-2](#); preprint arXiv:[2407.14610v1](#).